

TIME SERIES ECONOMETRICS

Empirical Applications in EViews

ARMA, VAR, Markov Switching
and Diebold–Li Models

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2026

Contents

1	European GDP Growth: Dynamic and Regime-Switching Analysis	2
1.1	ARMA modeling of growth rates	2
1.1.1	Italy	2
1.1.2	Spain	5
1.1.3	France	9
1.1.4	Finland	11
1.2	VAR models, Granger causality, and impulse-response functions	14
1.2.1	Italy – Spain	14
1.2.2	France – Finland	21
1.3	Markov regime-switching models	26
1.3.1	Italy	27
1.3.2	Spain	28
1.3.3	France	30
1.3.4	Finland	32
1.4	Business-cycle leadership	34
1.5	Comparison of causality and cyclical leadership results	35
1.6	Analysis over the Great Moderation period	36
1.6.1	ARMA modeling	37
1.6.2	VAR model	39
1.6.3	Markov regime-switching models	44
2	U.S. and U.K. Interest Rates: Term Structure and Dynamic Analysis	48
2.1	Stationarity of U.S. and U.K. interest rates	48
2.1.1	U.S. interest rates	48
2.1.2	U.K. interest rates	49
2.1.3	Interest rate differentials	51
2.1.4	Comparison of results	52
2.2	Cointegration analysis between interest rates	53
2.2.1	United States	53
2.2.2	United Kingdom	54
2.3	Implied devaluations and 3-month and 10-year rate differentials	55
2.4	Estimation of the Diebold-Li model via the Kalman filter, using the rate differential	57
2.5	Estimation of the Diebold-Li model via the Kalman filter, using implied devaluations	59
	Appendix A Full ADF test results	62
A.1	United States	62
A.2	United Kingdom	64
A.3	Rate differentials	66
	Appendix B Johansen test results	68

1 European GDP Growth: Dynamic and Regime-Switching Analysis

The dataset contains quarterly growth rates of real GDP for four European economies — France, Finland, Italy, and Spain — over the period from 1977Q1 to 2019Q4.

The objective of this analysis is to examine the dynamics of economic growth rates using different time series econometric tools. First, ARMA models are estimated to characterize the individual behavior of each economy. Subsequently, the dynamic relationships between countries are examined using VAR models and Granger causality tests. Markov-switching models are then employed to identify periods of expansion and recession, while also evaluating potential leadership relationships between business cycles. Finally, the results obtained for the two country pairs are compared, and the impact of the Great Moderation on the observed dynamics is analyzed for one of the country pairs.

1.1 ARMA modeling of growth rates

In order to characterize the temporal dynamics of real GDP growth rates for Italy, Spain, France, and Finland, a graphical analysis of the series was carried out first. The autocorrelation (ACF) and partial autocorrelation (PACF) functions were then examined through the corresponding correlograms, with the aim of identifying possible autoregressive and moving-average structures.

Based on the evidence provided by the correlograms, several candidate ARMA models were proposed for each economy. The final selection was made considering both the statistical fit of the models and the behavior of the residuals, verifying that they showed no significant evidence of remaining autocorrelation.

For Italy, AR(1) and AR(2) models were initially considered. After estimation and residual analysis, the AR(1) model proved sufficient to capture the temporal dependence observed in the series, and was therefore selected as the final specification.

For Spain, AR(2) and AR(3) models were initially estimated, but residual analysis suggested the persistence of serial correlation. Consequently, a moving-average component was added, and an ARMA(3,1) model was estimated, which showed better performance and residuals consistent with white noise.

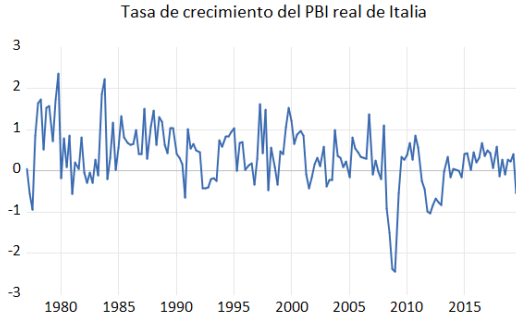
For France, AR(1) and AR(2) models were estimated. The results indicated that the AR(2) model represented the dynamics of the series more adequately, eliminating the residual serial dependence observed in more parsimonious specifications.

Finally, for Finland, AR(2) and AR(3) models were considered. Diagnostic analysis showed an improvement in fit when a third autoregressive lag was included, so the AR(3) model was adopted as the final specification.

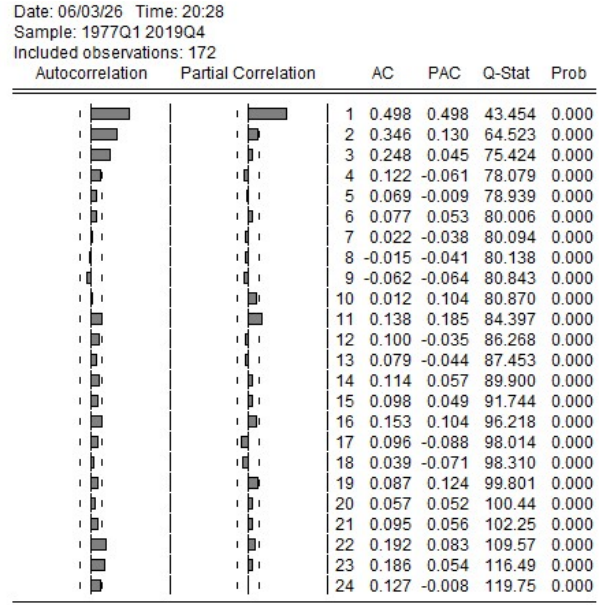
The selected models were subsequently subjected to diagnostic analysis through the study of their residuals, verifying that the relevant dynamic structure had been adequately captured.

1.1.1 Italy

Figure 1 presents the evolution of Italy's quarterly real GDP growth rate and its corresponding correlogram for the period 1977Q1–2019Q4.



(a) Italy's real GDP growth rate



(b) Correlogram for Italy

Figure 1: Real GDP growth rate and correlogram for Italy

Visual inspection of the series shows that Italian economic growth fluctuates around a relatively stable mean, although it exhibits episodes of high volatility associated with different macroeconomic events. The sharp contraction recorded during the 2008–2009 international financial crisis stands out in particular.

The correlogram shows significant positive autocorrelation in the first lags and a gradual decline in the autocorrelation function. The partial autocorrelation function, in turn, shows a clearly significant first lag and a rapid decline in subsequent lags. This pattern is consistent with a low-order autoregressive structure.

Based on this evidence, two alternative specifications were initially proposed: an AR(1) model and an AR(2) model.

Figure 2 presents the estimation of the AR(1) model together with the correlogram of its residuals. Both the constant and the autoregressive coefficient are statistically significant at the 1% level. The AR(1) parameter indicates moderate persistence in the dynamics of Italian economic growth, suggesting that roughly half of the variation observed in one quarter tends to carry over into the following quarter.

Residual analysis shows that the residual autocorrelations are not statistically significant and that the p-values associated with the Ljung-Box Q statistic remain above conventional significance levels for all lags considered. Therefore, there is no evidence of residual autocorrelation.

Date: 06/04/26 Time: 01:59
Sample: 1977Q1 2019Q4
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.064	-0.064	0.7186	
		2 0.084	0.080	1.9501	0.163
		3 0.102	0.114	3.8025	0.149
		4 -0.007	0.000	3.8101	0.283
		5 -0.017	-0.037	3.8624	0.425
		6 0.069	0.055	4.7139	0.452
		7 -0.004	0.010	4.7171	0.581
		8 0.004	-0.001	4.7199	0.694
		9 -0.103	-0.120	6.6578	0.574
		10 -0.028	-0.045	6.8046	0.657
		11 0.155	0.182	11.288	0.336
		12 0.021	0.075	11.368	0.413
		13 -0.009	-0.035	11.383	0.496
		14 0.069	0.012	12.290	0.504
		15 -0.013	0.007	12.323	0.580
		16 0.125	0.151	15.309	0.429
		17 0.032	0.021	15.507	0.488
		18 -0.056	-0.123	16.109	0.516
		19 0.081	0.037	17.404	0.496
		20 -0.028	0.044	17.558	0.552
		21 -0.005	0.037	17.564	0.616
		22 0.133	0.073	21.083	0.454
		23 0.098	0.094	23.001	0.402
		24 0.061	0.094	23.757	0.417

(a) Correlogram of the AR(1) residuals

Dependent Variable: ITALY
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/04/26 Time: 01:37
Sample: 1977Q1 2019Q4
Included observations: 172
Convergence achieved after 19 iterations
Coefficient covariance computed using outer product of gradients
d.f. adjustment for standard errors & covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.319167	0.094896	3.363341	0.0010
AR(1)	0.497830	0.044518	11.18260	0.0000
SIGMASQ	0.383586	0.038641	9.926836	0.0000
R-squared	0.249498	Mean dependent var		0.324134
Adjusted R-squared	0.240616	S.D. dependent var		0.717004
S.E. of regression	0.624816	Akaike info criterion		1.916226
Sum squared resid	65.97683	Schwarz criterion		1.971124
Log likelihood	-161.7954	Hannan-Quinn criter.		1.938499
F-statistic	28.09129	Durbin-Watson stat		2.126887
Prob(F-statistic)	0.000000			
Inverted AR Roots	.50			

(b) Estimated AR(1) equation

Figure 2: AR(1) model for Italy

Figure 3 presents the results obtained for the AR(2) model. The first autoregressive lag is statistically significant, while the second lag is only marginally significant ($p=0.053$). Including the second lag captures an additional fraction of the dynamics of the series, although its contribution is relatively limited.

As with the AR(1) model, residual analysis shows no evidence of remaining autocorrelation, indicating that the specification adequately captures the temporal dependence present in the series.

Date: 06/04/26 Time: 02:15
Sample: 1977Q1 2019Q4
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.004	-0.004	0.0023	
		2 -0.013	-0.013	0.0314	
		3 0.066	0.066	0.8109	0.368
		4 -0.037	-0.037	1.0610	0.588
		5 -0.038	-0.036	1.3133	0.726
		6 0.062	0.057	1.9976	0.736
		7 0.004	0.009	2.0010	0.849
		8 -0.018	-0.014	2.0602	0.914
		9 -0.130	-0.142	5.1713	0.639
		10 -0.032	-0.031	5.3552	0.719
		11 0.166	0.176	10.456	0.315
		12 0.029	0.046	10.613	0.388
		13 -0.021	-0.033	10.698	0.469
		14 0.047	0.011	11.123	0.518
		15 -0.007	0.019	11.133	0.600
		16 0.123	0.162	14.035	0.447
		17 0.025	-0.004	14.159	0.514
		18 -0.066	-0.112	15.001	0.525
		19 0.059	0.045	15.672	0.547
		20 -0.044	0.023	16.055	0.589
		21 -0.019	0.030	16.127	0.649
		22 0.138	0.075	19.936	0.462
		23 0.118	0.108	22.724	0.359
		24 0.048	0.093	23.198	0.391

(a) Correlogram of the AR(2) residuals

Dependent Variable: ITALY
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/04/26 Time: 01:48
Sample: 1977Q1 2019Q4
Included observations: 172
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.314533	0.108291	2.904517	0.0042
AR(1)	0.430891	0.055170	7.810296	0.0000
AR(2)	0.135272	0.069479	1.946949	0.0532
SIGMASQ	0.376533	0.037052	10.16230	0.0000
R-squared	0.263299	Mean dependent var	0.324134	
Adjusted R-squared	0.250143	S.D. dependent var	0.717004	
S.E. of regression	0.620885	Akaike info criterion	1.909512	
Sum squared resid	64.76360	Schwarz criterion	1.982710	
Log likelihood	-160.2180	Hannan-Quinn criter.	1.939210	
F-statistic	20.01453	Durbin-Watson stat	2.005546	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.64	-.21		

(b) Estimated AR(2) equation

Figure 3: AR(2) model for Italy

Comparing both specifications shows that the AR(2) model has a slightly lower Akaike criterion than the AR(1). However, the improvement in fit is small, and the coefficient on the second lag is only marginally significant. Since both models produce residuals consistent with white noise, and the AR(1) is a more parsimonious specification, the latter is selected as the final model to represent the dynamics of Italy's GDP growth rate.

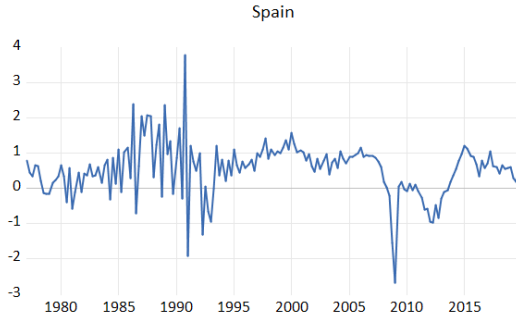
It is therefore concluded that the evolution of Italian economic growth can be adequately described by a first-order autoregressive process characterized by moderate persistence and stable dynamics over time.

1.1.2 Spain

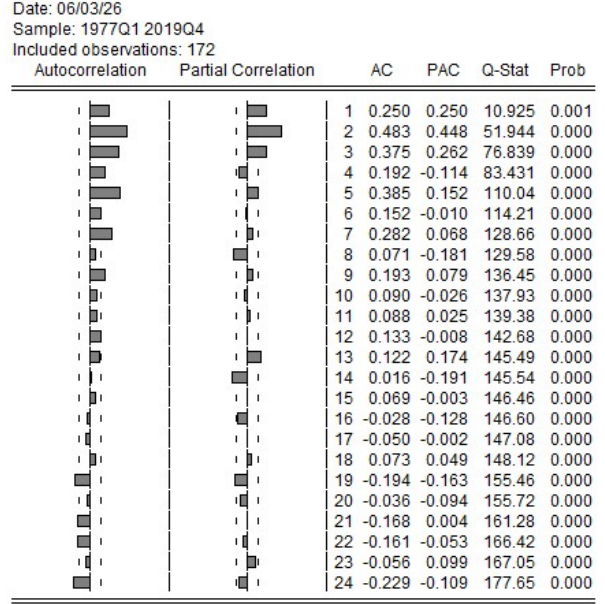
Figure 4 presents the evolution of Spain's quarterly real GDP growth rate for the period 1977Q1–2019Q4 and the corresponding correlogram.

Visually, the series shows fluctuations around a positive mean, although with significant episodes of volatility. The behavior observed between the late 1980s and early 1990s is particularly noteworthy, with variations of considerable magnitude. A sharp contraction is also observed during the 2008–2009 international financial crisis, followed by a period of gradual recovery.

The correlogram reveals a more complex dependence structure than that observed for Italy. The autocorrelation function shows significant positive values over several consecutive lags, while the partial autocorrelation function does not show a clear cutoff in the first lags. This behavior suggests the existence of persistent dynamics that may require higher-order models to be adequately represented.



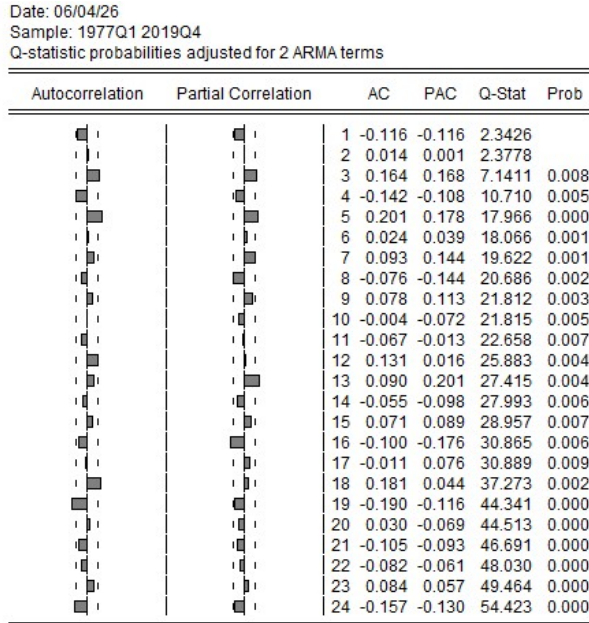
(a) Spain's real GDP growth rate



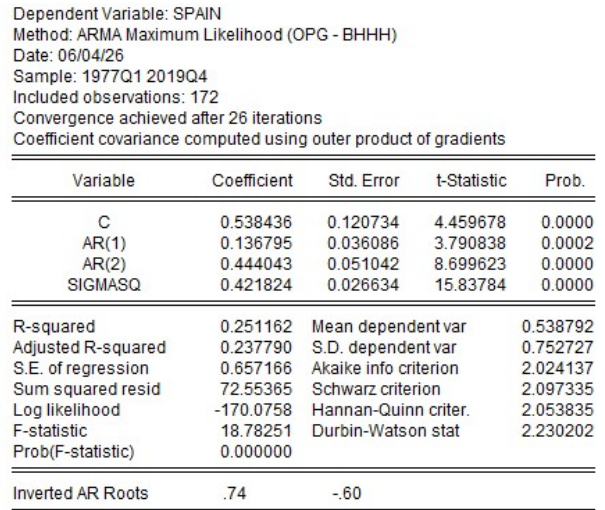
(b) Correlogram for Spain

Figure 4: Real GDP growth rate and correlogram for Spain

Based on this evidence, the first specification considered was an AR(2) model, whose results are presented in Figure 5.



(a) Correlogram of the AR(2) residuals



(b) Estimated AR(2) equation

Figure 5: AR(2) model for Spain

Both the constant and the coefficients associated with the first and second lags are statistically significant at

the 1% level. The coefficient corresponding to the second lag is considerably larger in magnitude than that of the first lag, indicating that part of the dynamics of Spanish economic growth manifests with some time delay.

However, residual analysis reveals that this specification does not fully capture the temporal dependence present in the series. The residual correlogram shows significant autocorrelations at several lags, and the p-values associated with the Ljung-Box Q statistic are below 5% for several horizons. Consequently, the null hypothesis of no residual autocorrelation is rejected.

These results suggest that the AR(2) model leaves systematic information unexplained and therefore does not constitute an adequate representation of the dynamics of the series.

In order to capture greater temporal persistence, an AR(3) model was subsequently estimated, adding an additional lag (Figure 6).

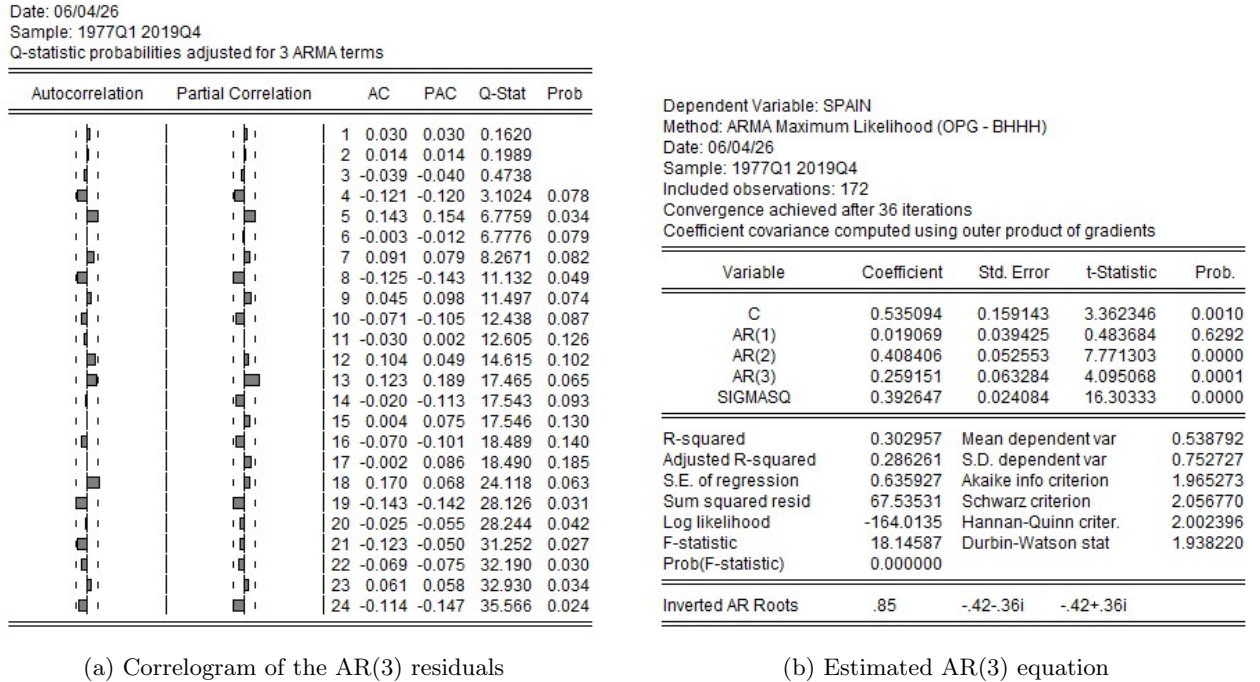


Figure 6: AR(3) model for Spain

The coefficients corresponding to the second and third lags are highly significant, while the first lag loses statistical significance. This result suggests that the dynamics of Spanish growth depend mainly on movements observed two and three quarters earlier.

Adding the third lag produces a substantial improvement in the model's fit. The Akaike criterion decreases from 2.024 in the AR(2) model to 1.965 in the AR(3) model, reflecting a significant improvement in the explanatory power of the specification.

Likewise, the residual correlogram shows a significant reduction in remaining autocorrelation. However, some lags with relatively low significance levels are still observed, suggesting that the temporal dependence has not been fully eliminated.

Since the AR(3) model still showed residual evidence of serial correlation, a first-order moving-average component was added, and an ARMA(3,1) model was estimated (Figure 7).

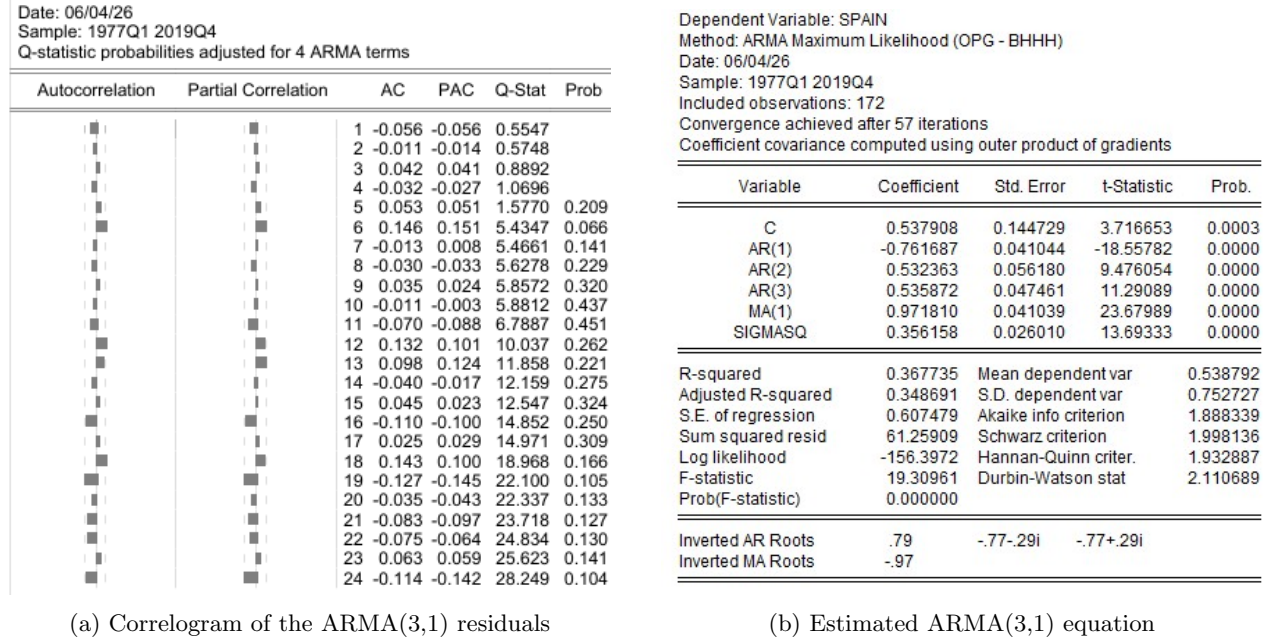


Figure 7: ARMA(3,1) model for Spain

All estimated parameters are highly statistically significant. In particular, the MA(1) coefficient is large in magnitude, indicating the presence of an important moving-average component in the dynamics of the series.

Adding this term substantially improves the overall fit of the model. The Akaike criterion falls to 1.888, the lowest value among all the specifications considered.

From a diagnostic standpoint, the residual correlogram shows that the remaining autocorrelations virtually disappear. The p-values associated with the Ljung-Box Q statistic remain above conventional significance levels for all lags analyzed, so there is no statistical evidence of residual autocorrelation.

Comparing the three specifications shows that the ARMA(3,1) model provides the best representation of the dynamics of Spain's GDP growth rate. While the AR(2) model shows significant residual autocorrelation and the AR(3) model only partially reduces it, adding the moving-average term effectively eliminates the remaining serial dependence.

Additionally, the ARMA(3,1) has the lowest value among the information criteria and produces residuals consistent with white noise. For these reasons, this specification is selected as the final model for subsequent analysis.

The simultaneous presence of several significant autoregressive components and a significant moving-average term indicates that both the past evolution of economic activity and recent shocks play a relevant role in determining current growth.

1.1.3 France

Figure 8 presents the evolution of France's real GDP growth rate and its corresponding correlogram.

Visual inspection of the series suggests the existence of temporal persistence in the French real GDP growth rate. Fluctuations associated with different macroeconomic episodes are also observed, notably the sharp decline recorded during the 2008–2009 international financial crisis and the subsequent recovery of economic activity.

The correlogram shows positive and statistically significant autocorrelations in the first lags. In particular, the autocorrelation function shows relevant values up to approximately the third lag and then declines rapidly, while the partial autocorrelation function concentrates its significance in the first lags and virtually disappears after the third. This pattern provides evidence in favor of a low-order autoregressive representation, which is why AR(1) and AR(2) specifications were initially estimated.

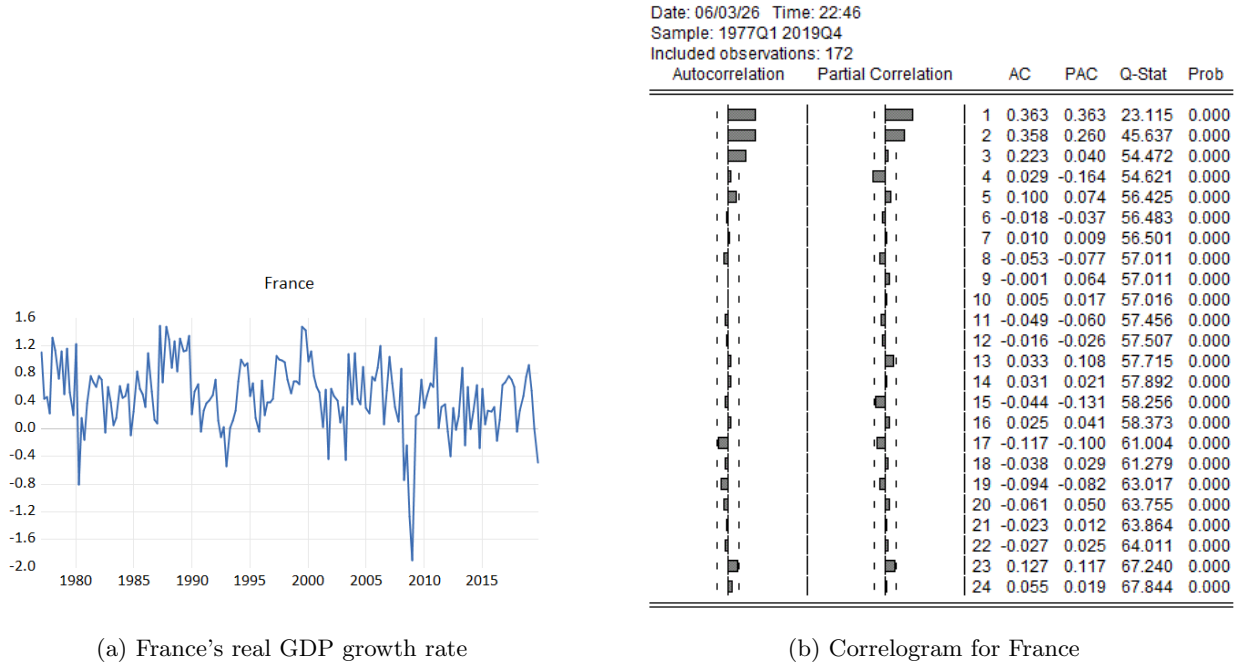


Figure 8: Real GDP growth rate and correlogram for France

The first specification estimated was a first-order autoregressive model (Figure 9). Both the constant and the autoregressive coefficient are statistically significant at the 1% level. The estimated parameter indicates moderate persistence in the dynamics of French economic growth.

However, analysis of the residual correlogram reveals the presence of remaining autocorrelation. The p-values associated with the Ljung-Box Q statistic are below conventional significance levels for numerous lags, indicating that the model does not fully capture the temporal dependence present in the series. For this reason, the AR(1) specification is considered insufficient.

Date: 06/04/26 Time: 04:28
Sample: 1977Q1 2019Q4
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.099	-0.099	1.7055	
		2 0.223	0.216	10.488	0.001
		3 0.129	0.177	13.432	0.001
		4 -0.106	-0.136	15.429	0.001
		5 0.124	0.035	18.203	0.001
		6 -0.072	-0.023	19.139	0.002
		7 0.043	0.032	19.475	0.003
		8 -0.068	-0.085	20.325	0.005
		9 0.019	0.025	20.389	0.009
		10 0.027	0.046	20.519	0.015
		11 -0.061	-0.037	21.211	0.020
		12 -0.018	-0.081	21.268	0.031
		13 0.037	0.073	21.523	0.043
		14 0.045	0.100	21.914	0.057
		15 -0.075	-0.106	22.995	0.060
		16 0.102	0.037	25.006	0.050
		17 -0.147	-0.100	29.194	0.023
		18 0.042	0.024	29.534	0.030
		19 -0.077	-0.085	30.707	0.031
		20 -0.034	0.008	30.934	0.041
		21 0.015	0.012	30.978	0.055
		22 -0.082	-0.018	32.322	0.054
		23 0.155	0.104	37.173	0.023
		24 -0.025	0.053	37.299	0.030

(a) Correlogram of the AR(1) residuals

Dependent Variable: FRANCE
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/04/26 Time: 04:27
Sample: 1977Q1 2019Q4
Included observations: 172
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.472421	0.060525	7.805407	0.0000
AR(1)	0.372594	0.051996	7.165754	0.0000
SIGMASQ	0.218822	0.019475	11.23605	0.0000

R-squared	0.136218	Mean dependent var	0.473555
Adjusted R-squared	0.125996	S.D. dependent var	0.504789
S.E. of regression	0.471918	Akaike info criterion	1.354133
Sum squared resid	37.63738	Schwarz criterion	1.409031
Log likelihood	-113.4554	Hannan-Quinn criter.	1.376407
F-statistic	13.32562	Durbin-Watson stat	2.171984
Prob(F-statistic)	0.000004		

Inverted AR Roots	.37
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(b) Estimated AR(1) equation

Figure 9: AR(1) model for France

In order to capture greater temporal persistence, an AR(2) model was subsequently estimated (Figure 10).

The coefficients associated with both lags are positive and statistically significant. Likewise, adding the second lag produces an improvement in the information criteria and allows for a more adequate representation of the dynamics observed in the series.

Residual analysis shows a substantial improvement relative to the previous model. The residual autocorrelations virtually disappear, and the p-values associated with the Ljung-Box Q statistic remain above conventional significance levels for all lags analyzed. Consequently, there is no evidence of residual autocorrelation.

Comparing both specifications shows that the AR(2) model provides the most adequate representation for France's GDP growth rate. While the AR(1) model leaves evidence of residual autocorrelation, adding a second lag more fully captures the temporal dependence present in the series. The AR(2) model also has better information criteria and produces residuals consistent with white noise.

For these reasons, the AR(2) model is selected as the final specification describing the dynamics of French GDP growth.

From an economic standpoint, the positive and significant autoregressive coefficients indicate that French economic growth has a degree of temporal inertia. This implies that economic conditions observed in one quarter influence the evolution of activity in subsequent periods. In particular, the model suggests that both growth recorded one quarter earlier and growth observed two quarters earlier help explain current growth. Shocks affecting the French economy are therefore not absorbed instantaneously; rather, their effects persist over several quarters before gradually dissipating.

Date: 06/04/26 Time: 04:24
Sample: 1977Q1 2019Q4
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1	-0.010	-0.010	0.0186
		2	0.036	0.036	0.2460
		3	0.073	0.074	1.1909
		4	-0.158	-0.159	5.6596
		5	0.056	0.050	6.2187
		6	-0.050	-0.045	6.6779
		7	-0.002	0.018	6.6786
		8	-0.066	-0.100	7.4826
		9	0.028	0.056	7.6257
		10	0.042	0.027	7.9518
		11	-0.081	-0.067	9.1685
		12	-0.035	-0.076	9.3992
		13	0.071	0.101	10.339
		14	0.042	0.058	10.677
		15	-0.042	-0.071	11.005
		16	0.081	0.056	12.276
		17	-0.118	-0.093	14.953
		18	0.013	0.029	14.986
		19	-0.060	-0.109	15.689
		20	-0.038	0.017	15.978
		21	-0.015	-0.043	16.022
		22	-0.060	-0.022	16.748
		23	0.144	0.089	20.905
		24	0.026	0.064	21.037

(a) Correlogram of the AR(2) residuals

Dependent Variable: FRANCE
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/04/26 Time: 04:21
Sample: 1977Q1 2019Q4
Included observations: 172
Convergence achieved after 24 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.469609	0.079070	5.939184	0.0000
AR(1)	0.274899	0.056910	4.830427	0.0000
AR(2)	0.266397	0.079889	3.334566	0.0011
SIGMASQ	0.203439	0.019220	10.58456	0.0000
R-squared	0.196941	Mean dependent var	0.473555	
Adjusted R-squared	0.182600	S.D. dependent var	0.504789	
S.E. of regression	0.456380	Akaike info criterion	1.293736	
Sum squared resid	34.99153	Schwarz criterion	1.366933	
Log likelihood	-107.2613	Hannan-Quinn criter.	1.323434	
F-statistic	13.73333	Durbin-Watson stat	1.990776	
Prob(F-statistic)	0.000000			
Inverted AR Roots	.67	-.40		

(b) Estimated AR(2) equation

Figure 10: AR(2) model for France

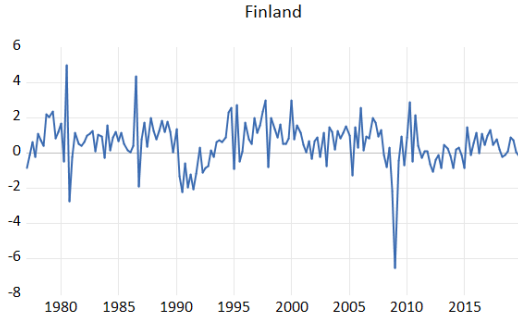
1.1.4 Finland

Figure 11 presents the evolution of Finland's quarterly real GDP growth rate over the period 1977Q1–2019Q4, and the corresponding correlogram.

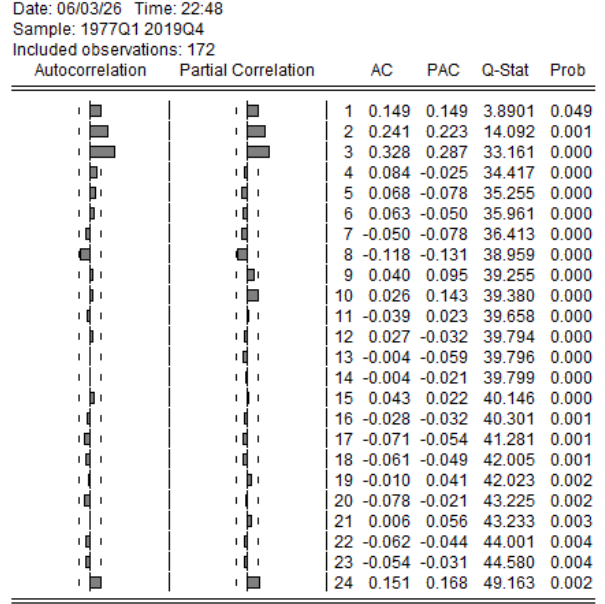
Visual inspection of the series shows that Finnish economic growth fluctuates around a relatively stable mean, without a clear deterministic trend over the period analyzed. Nevertheless, episodes of high volatility associated with different economic events are identified. In particular, the shocks observed during 1980–1981, 1986–1987, and especially the sharp contraction recorded during the 2008–2009 international financial crisis, stand out.

The correlogram shows positive and statistically significant autocorrelations in the first lags. The autocorrelation function reaches its highest values up to approximately the third lag, declining rapidly thereafter. The partial autocorrelation function likewise concentrates most of its significance in the first lags and virtually disappears after the third. This behavior is consistent with a low-order autoregressive structure.

Based on this evidence, AR(2) and AR(3) models were initially considered.



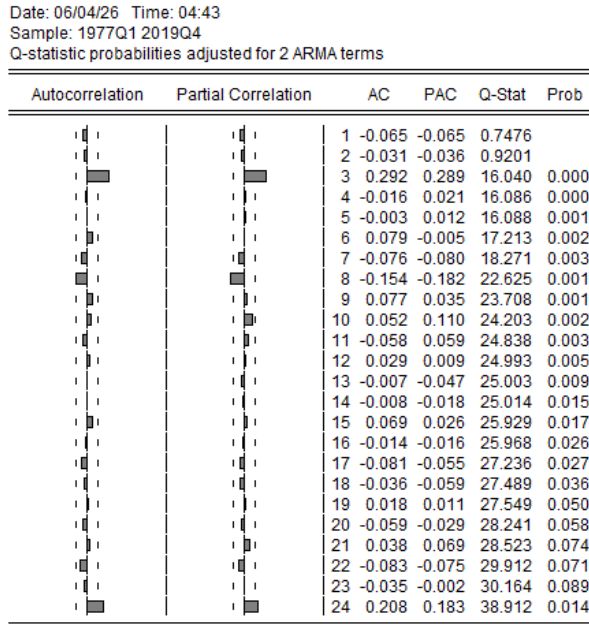
(a) Finland's real GDP growth rate



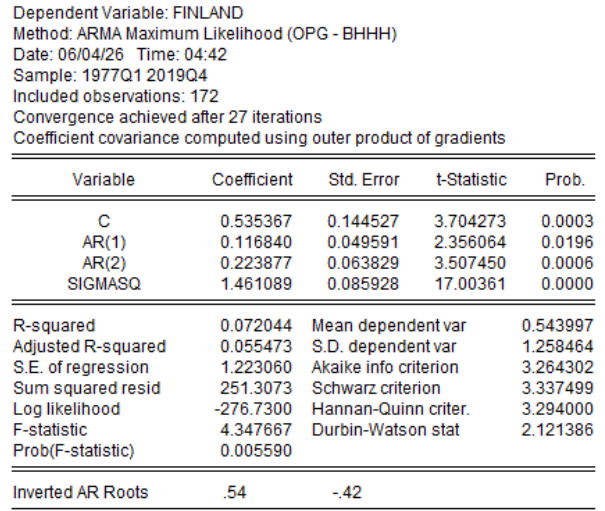
(b) Correlogram for Finland

Figure 11: Real GDP growth rate and correlogram for Finland

The first specification estimated was a second-order autoregressive model (Figure 12).



(a) Correlogram of the AR(2) residuals



(b) Estimated AR(2) equation

Figure 12: AR(2) model for Finland

Both the constant and the coefficients associated with the two lags are statistically significant at conventional levels. The estimated parameters suggest the existence of some temporal persistence in the dynamics of

Finnish economic growth.

However, residual analysis shows that this specification does not fully capture the temporal dependence present in the series. The residual correlogram shows several significant autocorrelations, and the p-values associated with the Ljung-Box Q statistic are below 5% for numerous lags. Consequently, the null hypothesis of no residual autocorrelation is rejected.

These results indicate that the AR(2) model leaves systematic information unexplained and therefore does not constitute an adequate representation of the series.

In order to capture greater temporal persistence, an AR(3) model was subsequently estimated (Figure 13).

The constant, as well as the coefficients corresponding to the second and third lags, are statistically significant. The coefficient associated with the first lag, on the other hand, is not individually significant. Nevertheless, adding the third lag produces a substantial improvement in the model's overall fit.

The information criteria show an appreciable reduction relative to the AR(2) model, with the Akaike criterion falling from 3.264 to 3.190. Residual analysis also shows a significant improvement. The residual autocorrelations virtually disappear, and the p-values of the Ljung-Box Q statistic remain above conventional significance levels for all lags analyzed.

Therefore, there is no evidence of residual autocorrelation, indicating that the AR(3) model adequately captures the temporal dynamics of the series.

Date: 06/04/26 Time: 04:48

Sample: 1977Q1 2019Q4

Q-statistic probabilities adjusted for 3 ARMA terms

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1	0.007	0.007	0.0074
		2	0.027	0.027	0.1404
		3	0.037	0.036	0.3774
		4	0.021	0.020	0.4544
		5	-0.007	-0.009	0.4635
		6	-0.033	-0.035	0.6570
		7	-0.123	-0.124	3.3923
		8	-0.181	-0.182	9.3710
		9	0.041	0.050	9.6867
		10	0.072	0.098	10.635
		11	-0.017	0.004	10.688
		12	0.002	-0.000	10.689
		13	-0.004	-0.025	10.693
		14	0.032	-0.000	10.883
		15	0.080	0.043	12.103
		16	-0.020	-0.036	12.181
		17	-0.074	-0.047	13.247
		18	-0.076	-0.061	14.384
		19	0.029	0.021	14.545
		20	-0.036	-0.031	14.796
		21	-0.010	0.003	14.814
		22	-0.042	-0.019	15.164
		23	-0.014	-0.002	15.204
		24	0.207	0.188	23.874

(a) Correlogram of the AR(3) residuals

Dependent Variable: FINLAND

Method: ARMA Maximum Likelihood (OPG - BHHH)

Date: 06/04/26 Time: 04:47

Sample: 1977Q1 2019Q4

Included observations: 172

Convergence achieved after 28 iterations

Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.524687	0.193189	2.715920	0.0073
AR(1)	0.051053	0.055143	0.925833	0.3559
AR(2)	0.191283	0.064311	2.974353	0.0034
AR(3)	0.286254	0.103674	2.761107	0.0064
SIGMASQ	1.338658	0.075942	17.62735	0.0000
R-squared	0.149801	Mean dependent var		0.543997
Adjusted R-squared	0.129437	S.D. dependent var		1.258464
S.E. of regression	1.174197	Akaike info criterion		3.189905
Sum squared resid	230.2492	Schwarz criterion		3.281402
Log likelihood	-269.3319	Hannan-Quinn criter.		3.227028
F-statistic	7.356147	Durbin-Watson stat		1.976615
Prob(F-statistic)	0.000018			
Inverted AR Roots	.77	-.36-.49i	-.36+.49i	

(b) Estimated AR(3) equation

Figure 13: AR(3) model for Finland

Comparing both specifications shows that the AR(3) model provides the most adequate representation for Finland's GDP growth rate. While the AR(2) model leaves clear evidence of residual autocorrelation, adding

a third lag eliminates this dependence and simultaneously improves the information criteria.

For these reasons, the AR(3) model is selected as the final specification describing the dynamics of Finnish economic growth.

From an economic standpoint, the results suggest that the evolution of economic activity in Finland shows persistence distributed over several quarters. The significance of the second and third lags indicates that economic conditions observed several periods earlier continue to influence current growth. Consequently, shocks affecting the Finnish economy are not absorbed immediately; rather, their effects are transmitted gradually over time before dissipating.

1.2 VAR models, Granger causality, and impulse-response functions

This section analyzes the joint dynamics of GDP growth rates for the assigned country pairs using Vector Autoregression (VAR) models. Unlike the univariate analysis carried out in the previous section, VAR models allow the simultaneous study of several endogenous variables, incorporating both the influence of their own lags and the lags of the other variables in the system.

The objective is to evaluate whether dynamic relationships exist between the economies considered, to determine the direction of temporal causality via Granger causality tests, and to analyze the propagation of shocks through impulse-response functions. VAR models are estimated for both assigned country pairs: Italy–Spain and France–Finland.

1.2.1 Italy – Spain

To analyze the dynamic interrelationships between Italy’s and Spain’s GDP growth rates, a Vector Autoregression (VAR) model was estimated. This approach allows both variables to be modeled jointly, considering that each may depend both on its own lags and on the lags of the other economy. This makes it possible to study the existence of dynamic relationships between the two countries, evaluate the direction of temporal causality, and analyze the propagation of shocks through the system.

Lag order selection

As a preliminary step before estimating the VAR model, the optimal number of lags was determined using the Akaike (AIC), Schwarz (SC), Hannan-Quinn (HQ), and final prediction error (FPE) information criteria. The results are presented in Figure 14.

The AIC, HQ, and FPE criteria select a model with three lags, while the Schwarz criterion favors a more parsimonious specification with a single lag. This discrepancy is common in practice, since different criteria penalize the inclusion of additional parameters differently.

In particular, the Schwarz criterion applies a more severe penalty for model complexity, while the Akaike criterion and the FPE tend to favor broader structures capable of capturing more dynamics.

VAR Lag Order Selection Criteria						
Endogenous variables: ITALY SPAIN						
Exogenous variables: C						
Date: 06/04/26 Time: 06:39						
Sample: 1977Q1 2019Q4						
Included observations: 166						
Lag	LogL	LR	FPE	AIC	SC	HQ
0	-356.0963	NA	0.256317	4.314413	4.351907	4.329632
1	-319.9927	70.90230	0.174100	3.927622	4.040104	3.973279
2	-305.3537	28.39597	0.153160	3.799443	3.986912	3.875538
3	-294.4504	20.88715*	0.140945*	3.716270*	3.978726*	3.822803*
4	-293.5265	1.747494	0.146283	3.753332	4.090776	3.890303
5	-290.0409	6.509395	0.147219	3.759529	4.171961	3.926938
6	-288.7984	2.290273	0.152231	3.792752	4.280172	3.990599

* indicates lag order selected by the criterion
 LR: sequential modified LR test statistic (each test at 5% level)
 FPE: Final prediction error
 AIC: Akaike information criterion
 SC: Schwarz information criterion
 HQ: Hannan-Quinn information criterion

Figure 14: Lag-order selection criteria for the Italy–Spain VAR

In this case, the evidence provided by the different criteria mostly favors including three lags. Furthermore, since the objective of the analysis is to study the interrelationships between the Italian and Spanish economies, it is reasonable to allow the effects of one variable to be transmitted over several periods. For this reason, a VAR(3) was estimated, considering that this specification offers an adequate balance between explanatory power and parsimony.

Estimation of the VAR(3) model

Once the optimal number of lags was determined, a VAR(3) model was estimated for the GDP growth rates of Italy and Spain. The estimation results are presented in Figure 15.

The estimated model allows for the simultaneous analysis of the dynamics of both economies, considering both the influence of their own lags and the lags of the other variable in the system.

In the equation for Italy, the coefficient associated with the first lag of the variable itself is positive and statistically significant, evidencing temporal persistence in Italian economic growth. The first lag of Spain is also statistically significant, indicating that the recent evolution of the Spanish economy provides relevant information for explaining Italy's subsequent dynamics.

Vector Autoregression Estimates		
Date: 06/04/26 Time: 06:41		
Sample (adjusted): 1977Q4 2019Q4		
Included observations: 169 after adjustments		
Standard errors in () & t-statistics in []		
	ITALY	SPAIN
ITALY(-1)	0.366240 (0.07891) [4.64096]	0.199178 (0.08045) [2.47566]
ITALY(-2)	0.078082 (0.08453) [0.92373]	0.106900 (0.08618) [1.24045]
ITALY(-3)	0.014646 (0.07919) [0.18494]	-0.210020 (0.08074) [-2.60120]
SPAIN(-1)	0.228669 (0.07411) [3.08572]	-0.016813 (0.07555) [-0.22254]
SPAIN(-2)	0.025910 (0.07010) [0.36962]	0.390328 (0.07147) [5.46168]
SPAIN(-3)	-0.054322 (0.07341) [-0.74001]	0.270684 (0.07484) [3.61690]
C	0.077541 (0.06488) [1.19510]	0.158275 (0.06615) [2.39273]
R-squared	0.302074	0.359597
Adj. R-squared	0.276225	0.335879
Sum sq. resids	59.62528	61.97445
S.E. equation	0.606677	0.618513
F-statistic	11.68603	15.16096
Log likelihood	-151.7669	-155.0322
Akaike AIC	1.878898	1.917541
Schwarz SC	2.008539	2.047181
Mean dependent	0.338484	0.539018
S.D. dependent	0.713109	0.758971
Determinant resid covariance (dof adj.)		0.136101
Determinant resid covariance		0.125060
Log likelihood		-303.9289
Akaike information criterion		3.762472
Schwarz criterion		4.021753
Number of coefficients		14

Figure 15: VAR(3) for Italy and Spain

On the other hand, in the equation for Spain, the second and third lags of the variable itself are statistically significant, reflecting a strong dependence on its own past behavior. In addition, the second lag of Italy is significant, suggesting that the evolution of the Italian economy also influences Spanish growth.

Overall, the results show that the growth rates of both countries exhibit temporal persistence and that cross effects exist between the economies. This preliminary evidence suggests the presence of dynamic interrelationships between Italy and Spain, an issue that will be examined in greater detail through Granger causality tests and impulse-response functions.

Stability analysis

Before proceeding with the causality analysis and impulse-response functions, it is necessary to verify that the estimated VAR model satisfies the stability condition. This condition guarantees that the effects of shocks on the system's variables are transitory and that responses converge to their long-run equilibrium value.

Figure 16 presents the stability analysis of the estimated VAR(3) for Italy and Spain. The left-hand panel reports the roots of the characteristic polynomial together with their moduli, while the right-hand panel shows the graphical representation of the inverse roots.

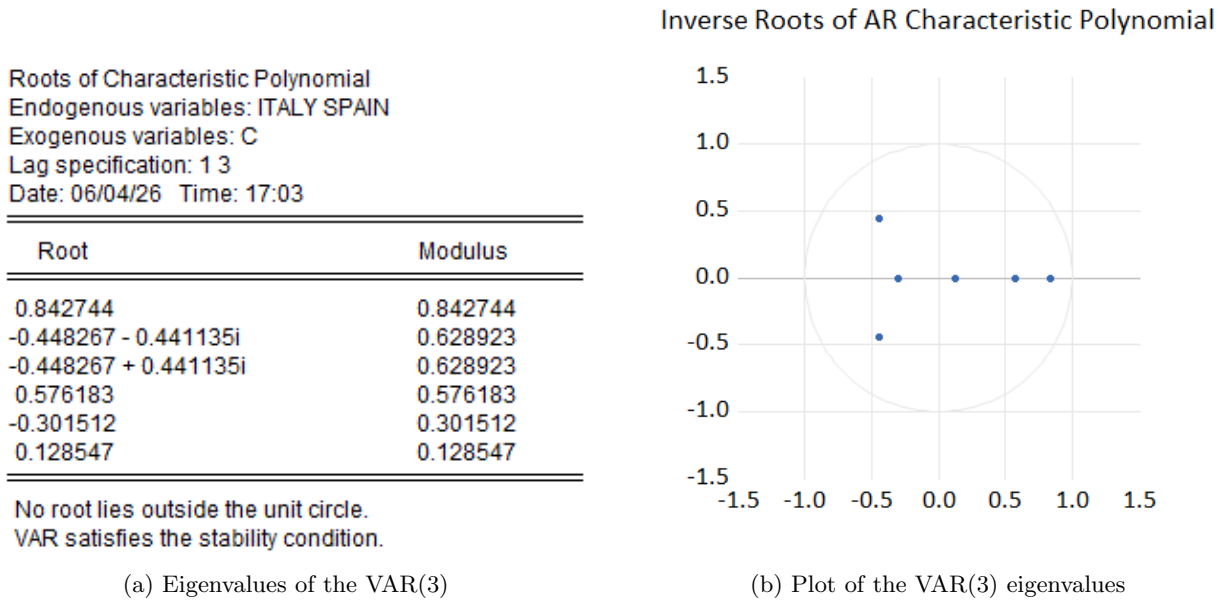


Figure 16: Stability analysis of the VAR(3) system for Italy and Spain

As shown in Figure 16(a), all roots of the characteristic polynomial have moduli below one. Likewise, the graphical representation of the inverse roots in Figure 16(b) shows that they lie entirely within the unit circle. The stability condition of the system is therefore verified.

The stability of the VAR implies that transitory shocks to the growth rates of Italy or Spain do not generate explosive trajectories in the system. On the contrary, the effects of such shocks tend to dissipate gradually over time. This property is essential for the validity of the subsequent Granger causality and impulse-response analyses.

Granger causality test

Once the stability of the VAR(3) model was verified, the existence of Granger causality relationships between the GDP growth rates of Italy and Spain was analyzed. This test evaluates whether information contained in the past values of one variable helps improve the prediction of another variable, once its own lags are taken into account.

VAR Granger Causality/Block Exogeneity Wald Tests
Date: 06/04/26 Time: 06:44
Sample: 1977Q1 2019Q4
Included observations: 169

Dependent variable: ITALY

Excluded	Chi-sq	df	Prob.
SPAIN	9.986542	3	0.0187
All	9.986542	3	0.0187

Dependent variable: SPAIN

Excluded	Chi-sq	df	Prob.
ITALY	14.13155	3	0.0027
All	14.13155	3	0.0027

Figure 17: Granger causality test for Italy and Spain

The results are presented in Figure 17. First, the null hypothesis that the lags of Spain's GDP growth rate do not Granger-cause Italy's GDP growth rate is tested. This hypothesis implies that the coefficients associated with Spain's lags in Italy's equation are jointly equal to zero. The resulting p-value is 0.0187, so the null hypothesis is rejected at the 5% significance level. There is therefore evidence that information contained in the past values of Spanish economic growth helps explain the future evolution of Italian growth.

Similarly, the null hypothesis that the lags of Italy's GDP growth rate do not Granger-cause Spain's GDP growth rate is evaluated. The p-value associated with this test is 0.0027, again rejecting the null hypothesis at the 1% significance level. There is therefore evidence that past values of Italian growth provide relevant information for predicting future Spanish growth.

Consequently, the results indicate bidirectional Granger causality between the GDP growth rates of Italy and Spain. This suggests that the past evolution of each economy contains useful information for anticipating the future behavior of the other, evidencing significant dynamic interdependence between the two countries.

From an economic standpoint, this result is consistent with the high degree of integration between the Italian and Spanish economies, both because of their membership in the European Union and because of the trade, financial, and institutional ties they maintain with each other. In this context, disturbances affecting economic activity in one country can transmit their effects to the other, generating dynamic relationships reflected in the causality results obtained.

Impulse-response functions

In order to analyze the dynamic transmission of disturbances between the Italian and Spanish economies, impulse-response functions (IRFs) were estimated based on the VAR(3) model.

The IRFs reported by EViews are based on orthogonalized innovations obtained through a Cholesky decomposition of the residual covariance matrix. This procedure transforms the system so that innovations are

independent of one another, resolving the correlation between the errors of the VAR equations. This makes it possible to analyze the dynamic effect of a shock to one variable without that effect being contaminated by simultaneous innovations in the other variables of the system.

However, an important feature of the Cholesky decomposition is that the results depend on the ordering of the variables within the model. For this reason, impulse-response functions are reported for both possible orderings, Italy-Spain and Spain-Italy, in order to evaluate the robustness of the results to changes in shock identification.

Figures 18 and 19 present the impulse-response functions obtained for the Italy-Spain and Spain-Italy orderings, respectively. In both graphs, the solid line represents the estimated response to a one-standard-deviation innovation, while the dashed lines correspond to approximate confidence bands of ± 2 standard errors.

As shown in both figures, own innovations generate the largest responses. For both Italy and Spain, a shock to the GDP growth rate produces a positive and immediate effect on the economy itself, reaching its maximum intensity in the first periods and gradually dissipating over time until converging to zero. This behavior is consistent with the stability condition previously verified for the VAR(3).

Cross responses, on the other hand, show evidence of shock transmission between the two economies. In particular, an innovation in Spain's GDP growth rate generates a positive response in Italy's growth, while an innovation in the Italian economy produces a favorable effect on Spanish growth. However, the magnitude of these effects is smaller than that observed for own shocks, suggesting that international shock transmission exists, though with moderate intensity.

Likewise, comparing both orderings indicates that the main conclusions remain virtually unchanged. While minor differences in the magnitude of some responses are observed, the signs, persistence, and general pattern of shock transmission are similar in both cases. This suggests that the results obtained are relatively robust to the ordering used in the Cholesky decomposition.

Finally, the impulse-response functions are consistent with the results obtained through the Granger causality test. While that test showed a bidirectional relationship between the two economies, the IRFs make it possible to visualize how these interactions materialize over time. Taken together, both analyses suggest that the Italian and Spanish economies exhibit closely linked dynamics, where disturbances affecting one of them tend to transmit to the other, although with lower intensity than the effects observed internally.

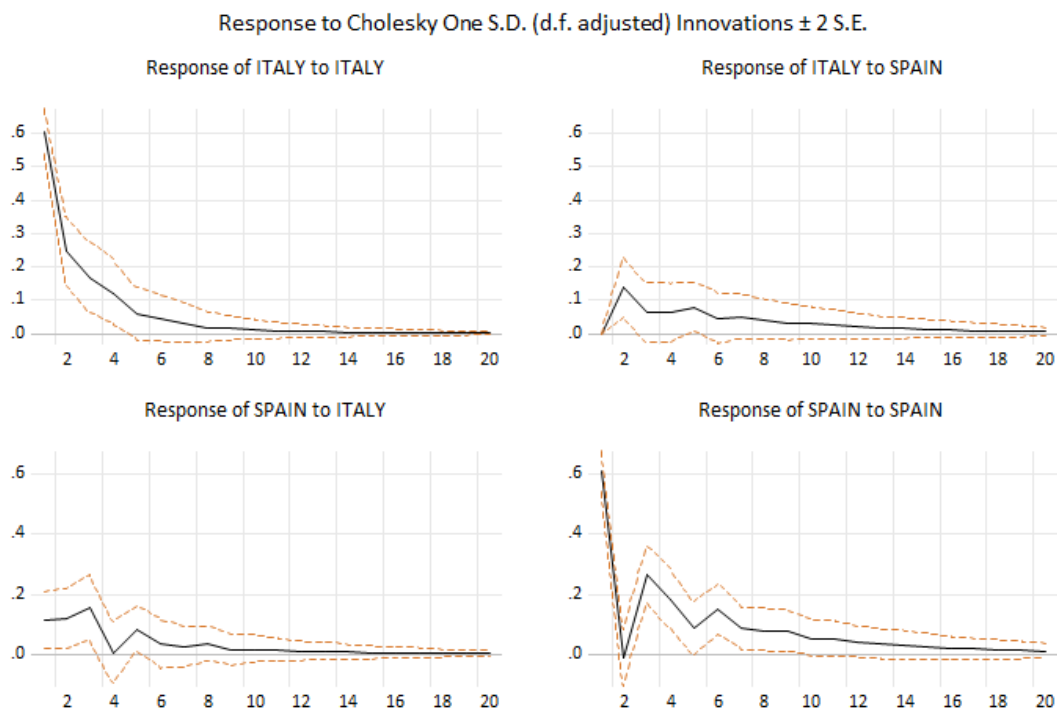


Figure 18: Impulse-response functions with Italy-Spain ordering

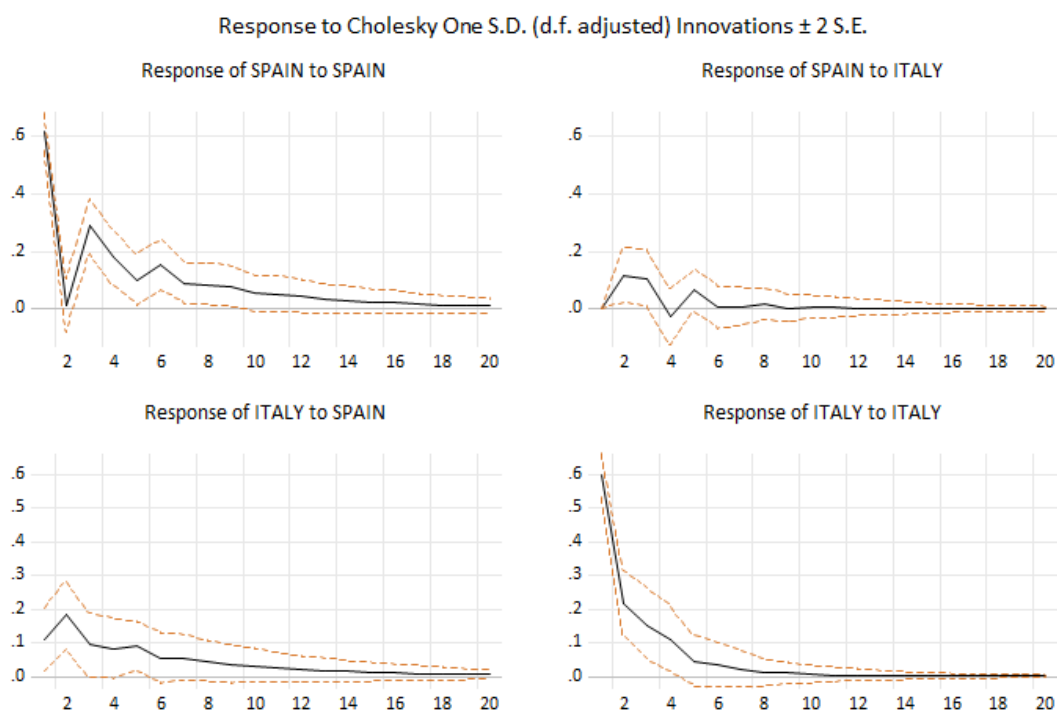


Figure 19: Impulse-response functions with Spain-Italy ordering

1.2.2 France – Finland

To analyze the dynamic interrelationships between France's and Finland's GDP growth rates, a Vector Autoregression (VAR) model was estimated. This approach allows both variables to be modeled jointly, considering that each may depend both on its own lags and on the lags of the other economy. This makes it possible to study the existence of dynamic relationships between the two countries, identify possible transmission mechanisms between their economies, and analyze the temporal propagation of shocks through the system.

Lag order selection

As a preliminary step before estimating the VAR model, the optimal number of lags was determined using the Akaike (AIC), Schwarz (SC), Hannan-Quinn (HQ), final prediction error (FPE), and sequential likelihood ratio (LR) test criteria. The results are presented in Figure 20.

The LR, FPE, and AIC criteria favor a specification with four lags, while the Schwarz criterion selects a model with no lags. The Hannan-Quinn criterion, in turn, selects a specification with three lags.

When information criteria do not coincide, the choice of the number of lags depends on the objective of the analysis. In general, the Akaike criterion tends to be preferred when the primary interest is prediction, while the Schwarz and Hannan-Quinn criteria tend to favor more parsimonious specifications and are usually preferred for inference exercises. Moreover, for relatively large sample sizes, these criteria are generally considered more reliable for selecting the VAR order.

Since the objective of this work is to analyze the dynamic relationships between the variables and perform inference on those relationships, the Hannan-Quinn criterion was followed. For these reasons, a VAR(3) model was estimated to study the interrelationships between the GDP growth rates of France and Finland.

VAR Lag Order Selection Criteria
Endogenous variables: FRANCE FINLAND
Exogenous variables: C
Date: 06/04/26 Time: 17:48
Sample: 1977Q1 2019Q4
Included observations: 166

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-371.7994	NA	0.309701	4.503607	4.541100	4.518826
1	-354.8582	33.26989	0.264992	4.347690	4.460171*	4.393347
2	-345.8733	17.42864	0.249552	4.287630	4.475099	4.363725
3	-336.8182	17.34656	0.234822	4.226725	4.489182	4.333258*
4	-331.0889	10.83723*	0.230006*	4.205891*	4.543335	4.342862
5	-329.3226	3.298634	0.236321	4.232802	4.645234	4.400211
6	-328.5261	1.468208	0.245683	4.271399	4.758818	4.469246

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Figure 20: Lag-order selection criteria for the France–Finland VAR

Estimation of the VAR(3) model

Once the optimal number of lags was determined, a VAR(3) model was estimated for the GDP growth rates of France and Finland. The estimation results are presented in Figure 21.

Vector Autoregression Estimates
Date: 06/05/26 Time: 02:53
Sample (adjusted): 1977Q4 2019Q4
Included observations: 169 after adjustments
Standard errors in () & t-statistics in []

	FRANCE	FINLAND
FRANCE(-1)	0.267831 (0.08669) [3.08970]	0.793533 (0.21478) [3.69459]
FRANCE(-2)	0.275859 (0.08769) [3.14569]	0.332962 (0.21728) [1.53239]
FRANCE(-3)	-0.023957 (0.08786) [-0.27265]	-0.427957 (0.21770) [-1.96578]
FINLAND(-1)	0.019435 (0.03323) [0.58481]	-0.101880 (0.08234) [-1.23728]
FINLAND(-2)	-0.040225 (0.03285) [-1.22439]	0.105877 (0.08140) [1.30069]
FINLAND(-3)	0.057842 (0.03247) [1.78139]	0.341806 (0.08045) [4.24857]
C	0.202077 (0.05714) [3.53658]	0.034910 (0.14158) [0.24658]
R-squared	0.228172	0.237798
Adj. R-squared	0.199586	0.209568
Sum sq. resids	33.31964	204.5545
S.E. equation	0.453516	1.123691
F-statistic	7.981890	8.423668
Log likelihood	-102.5936	-255.9347
Akaike AIC	1.296966	3.111653
Schwarz SC	1.426607	3.241294
Mean dependent	0.470135	0.555804
S.D. dependent	0.506915	1.263906
Determinant resid covariance (dof adj.)		0.216879
Determinant resid covariance		0.199285
Log likelihood		-343.3011
Akaike information criterion		4.228415
Schwarz criterion		4.487697
Number of coefficients		14

Figure 21: VAR(3) for France and Finland

In the equation for France, the coefficients associated with the first and second lags of the variable itself are statistically significant. This indicates that French economic growth exhibits temporal persistence and that its current evolution depends mainly on its own behavior in previous periods. By contrast, none of Finland's lags are statistically significant in this equation, suggesting a limited influence of Finnish growth on France's dynamics.

On the other hand, in the equation for Finland, the first lag of France and the third lag of Finland have statistically significant coefficients. These results suggest that both Finland's internal dynamics and France's past evolution contribute to explaining Finnish economic growth.

Overall, the VAR(3) estimation shows an asymmetric interaction between the two economies. While French growth appears to be determined mainly by internal factors, Finland's dynamics show greater sensitivity to France's past evolution. This preliminary evidence suggests a possible dependence relationship between the two economies, an issue analyzed further through Granger causality tests and impulse-response functions.

Stability analysis

A key condition for the validity of subsequent analyses is that the estimated VAR system is stable. To verify this condition, the roots of the characteristic polynomial associated with the model were analyzed, with results presented in Figure 22.

As shown in Figure 22(a), none of the roots has a modulus greater than one. In particular, the largest modulus is 0.7748, clearly below one. Likewise, the graphical representation of the inverse roots shown in Figure 22(b) indicates that all of them lie within the unit circle.

The stability condition of the VAR(3) is therefore verified, which implies that the effects of disturbances tend to dissipate gradually over time and that the system converges to its long-run behavior. This condition allows the results of the Granger causality tests and impulse-response functions presented in the following sections to be interpreted validly.

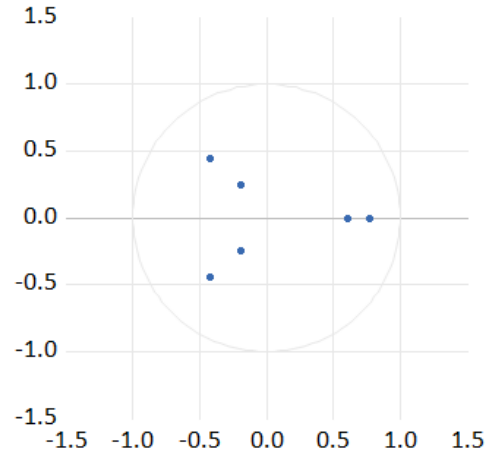
Roots of Characteristic Polynomial
Endogenous variables: FRANCE FINLAND
Exogenous variables: C
Lag specification: 1 3
Date: 06/05/26 Time: 03:19

Root	Modulus
0.774844	0.774844
-0.416869 - 0.445180i	0.609889
-0.416869 + 0.445180i	0.609889
0.606888	0.606888
-0.191021 - 0.241282i	0.307743
-0.191021 + 0.241282i	0.307743

No root lies outside the unit circle.
VAR satisfies the stability condition.

(a) Eigenvalues of the VAR(3)

Inverse Roots of AR Characteristic Polynomial



(b) Plot of the VAR(3) eigenvalues

Figure 22: Stability analysis of the VAR(3) system for France and Finland

Granger causality test

Once the stability of the VAR(3) model was verified, the existence of Granger causality relationships between the GDP growth rates of France and Finland was analyzed. This test evaluates whether information contained in the past values of one variable helps improve the prediction of another variable, once its own lags are taken into account.

The results are presented in Figure 23.

First, the null hypothesis that the lags of Finland's GDP growth rate do not Granger-cause France's GDP growth rate is tested. This hypothesis implies that the coefficients associated with Finland's lags in France's equation are jointly equal to zero. The resulting p-value is 0.1194, so the null hypothesis is not rejected at conventional significance levels. There is therefore not enough statistical evidence to conclude that the lags of Finland's GDP growth rate Granger-cause France's GDP growth rate.

Similarly, the null hypothesis that the lags of France's GDP growth rate do not Granger-cause Finland's GDP growth rate is evaluated. The p-value associated with this test is 0.0003, rejecting the null hypothesis at the 1% significance level. There is therefore evidence that the lags of France's GDP growth rate Granger-cause Finland's GDP growth rate.

Consequently, the results indicate a unidirectional Granger causality relationship running from France to Finland. This suggests that the past dynamics of the French economy contain useful information for anticipating the future evolution of the Finnish economy, while no evidence of predictive influence in the opposite direction is found.

From an economic standpoint, this result is consistent with France's larger relative size and greater economic weight within the European Union. In this context, changes in French economic activity can transmit to smaller, more open economies such as Finland's, while fluctuations in Finland do not appear to provide additional information for explaining the future behavior of the French economy.

VAR Granger Causality/Block Exogeneity Wald Tests			
Date: 06/05/26 Time: 02:57			
Sample: 1977Q1 2019Q4			
Included observations: 169			
Dependent variable: FRANCE			
Excluded	Chi-sq	df	Prob.
FINLAND	5.844373	3	0.1194
All	5.844373	3	0.1194
Dependent variable: FINLAND			
Excluded	Chi-sq	df	Prob.
FRANCE	18.71879	3	0.0003
All	18.71879	3	0.0003

Figure 23: Granger causality test for France and Finland

Impulse-response functions

In order to analyze the dynamic transmission of disturbances between the French and Finnish economies, impulse-response functions (IRFs) were estimated based on the VAR(3) model.

As in the case of Italy and Spain, impulse-response functions are reported for both possible variable order-

ings. This is because EViews computes IRFs from orthogonalized innovations obtained through a Cholesky decomposition, so the results may depend on the ordering used to identify the shocks.

Figures 24 and 25 present the impulse-response functions obtained for the France-Finland and Finland-France orderings, respectively. In both graphs, the solid line represents the estimated response to a one-standard-deviation orthogonalized innovation, while the dashed lines correspond to approximate confidence bands of ± 2 standard errors.

As shown in both figures, own innovations generate the largest responses. For both France and Finland, a shock to the GDP growth rate produces an immediate effect on the economy itself, reaching its maximum intensity in the first periods and gradually dissipating over time until converging to zero. This behavior is consistent with the stability condition previously verified for the VAR(3).

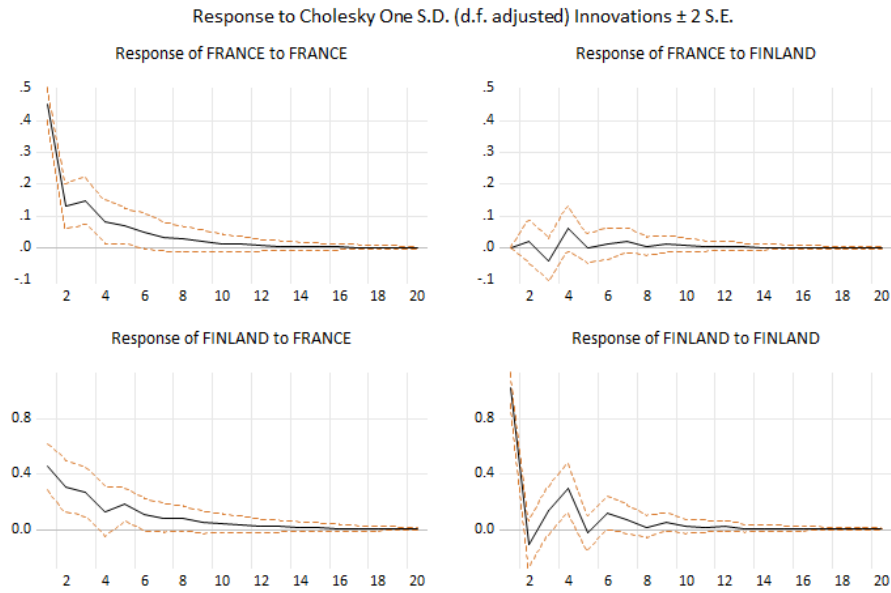


Figure 24: Impulse-response functions with France-Finland ordering

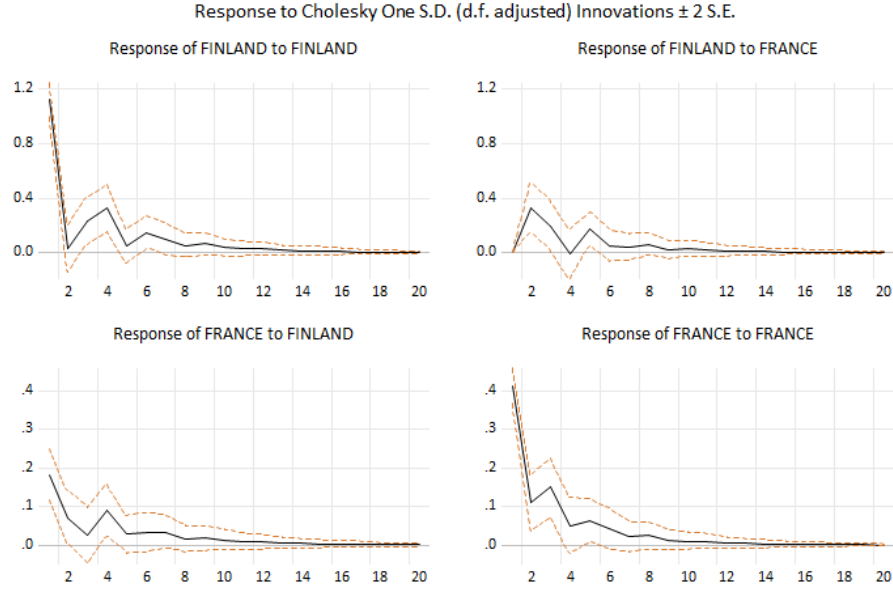


Figure 25: Impulse-response functions with Finland-France ordering

Cross responses, on the other hand, show an asymmetric transmission of disturbances between the two economies. In particular, an innovation in France's GDP growth rate generates a positive response in Finnish growth, whose effect persists over several periods before gradually disappearing. By contrast, France's response to an innovation originating in Finland is considerably smaller and oscillates around zero for most of the horizon analyzed.

Likewise, comparing both orderings shows that the general conclusions remain virtually unchanged. While minor differences in the magnitude of some responses are observed, the pattern of shock transmission remains stable, providing evidence of robustness to the ordering used in the Cholesky decomposition.

The impulse-response functions are consistent with the results obtained through the Granger causality test. While that test showed a unidirectional causality relationship running from France to Finland, the IRFs make it possible to visualize how a disturbance originating in the French economy generates appreciable effects on the Finnish economy. By contrast, shocks originating in Finland produce much smaller responses in France.

Taken together, the results suggest that the French economy exerts a relatively greater influence on Finland's growth dynamics than is observed in the opposite direction. This is consistent with France's greater size and economic weight within the European Union, as well as with the Granger causality result obtained previously.

1.3 Markov regime-switching models

In order to identify possible changes in the behavior of GDP growth rates over time, a two-state Markov regime-switching model was estimated for each country. In this specification, the mean of the series is allowed to depend on the prevailing regime, while the variance remains constant across states.

This type of model makes it possible to capture structural changes in economic dynamics and to classify each

period according to the prevailing regime. States are identified using the smoothed probabilities estimated by the model, which indicate the probability that the economy is in each regime at each point in time.

1.3.1 Italy

To identify possible regime changes in Italy's GDP growth rate, a two-state Markov regime-switching model was estimated, allowing only the mean to differ between regimes while the variance remains constant. The estimation results are presented in Figure 26.

The estimated parameters for both regimes are statistically significant at the 1% level. In particular, regime 1 has an estimated mean of -1.076, while regime 2 has a mean of 0.431. The difference between the two means suggests the existence of two clearly differentiated states in the dynamics of Italian economic growth.

From an economic standpoint, regime 1 can be interpreted as a recessionary state, given that the average growth rate is negative. Regime 2, in turn, can be interpreted as an expansionary regime, given that it exhibits a positive average growth rate. The model also incorporates a single common variance for both states, consistent with the proposed specification.

To identify the timing of the different regimes, the smoothed probabilities estimated by the model were analyzed; these are presented in Figure 27.

The smoothed probabilities show that regime 2 dominates for most of the period analyzed, with probabilities close to one from the late 1970s to approximately 2008, and again from 2014 to the end of the sample. This indicates that the Italian economy remained in an expansionary regime, characterized by a positive average growth rate, for most of the time.

By contrast, regime 1 appears concentrated in specific episodes. The most clearly identified periods correspond to the 2008–2009 international financial crisis and the 2011–2013 European sovereign debt crisis, during which the probability of being in the recessionary regime reaches values close to one. Additionally, isolated episodes of shorter duration are observed in the late 1970s and around the early 1990s, though with considerably lower intensity.

Dependent Variable: ITALY
Method: Markov Switching Regression (BFGS / Marquardt steps)
Date: 06/05/26 Time: 22:11
Sample: 1977Q1 2019Q4
Included observations: 172
Number of states: 2
Initial probabilities obtained from ergodic solution
Standard errors & covariance computed using observed Hessian
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=515477266)
Convergence achieved after 4 iterations

Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	-1.076145	0.215409	-4.995815	0.0000
Regime 2				
C	0.431484	0.049719	8.678434	0.0000
Common				
LOG(SIGMA)	-0.509734	0.056130	-9.081363	0.0000
Transition Matrix Parameters				
P11-C	1.441158	0.776506	1.855954	0.0635
P21-C	-4.219270	0.737321	-5.722431	0.0000
Mean dependent var	0.324134	S.D. dependent var	0.717004	
S.E. of regression	0.652216	Sum squared resid	71.89026	
Durbin-Watson stat	1.272746	Log likelihood	-169.3136	
Akaike info criterion	2.026903	Schwarz criterion	2.118399	
Hannan-Quinn criter.	2.064025			

Figure 26: Markov-switching model estimation for Italy

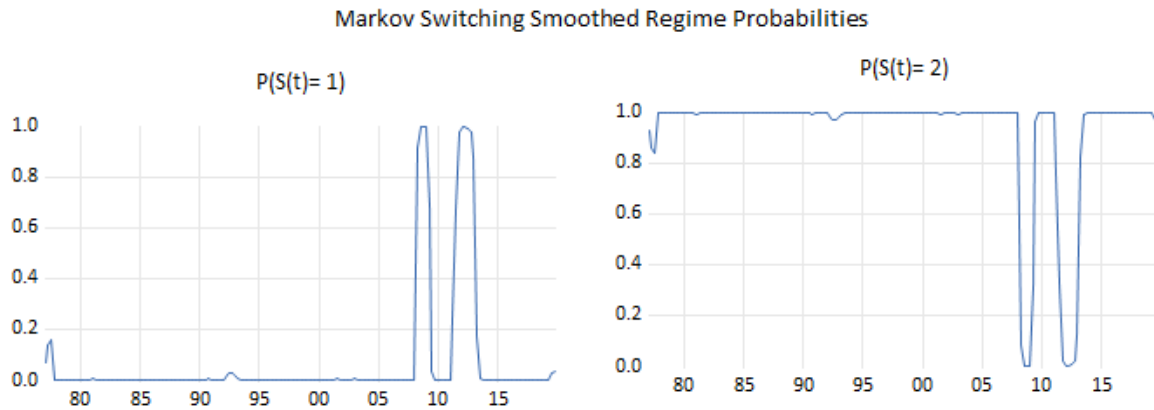


Figure 27: Smoothed regime probabilities for Italy

1.3.2 Spain

To identify possible regime changes in Spain's GDP growth rate, a two-state Markov regime-switching model was estimated, allowing only the mean to differ between regimes while the variance remains constant. The estimation results are presented in Figure 28.

Dependent Variable: SPAIN
Method: Markov Switching Regression (BFGS / Marquardt steps)
Date: 06/06/26 Time: 08:05
Sample: 1977Q1 2019Q4
Included observations: 172
Number of states: 2
Initial probabilities obtained from ergodic solution
Standard errors & covariance computed using observed Hessian
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=943195916)
Convergence achieved after 3 iterations

Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	0.802589	0.064883	12.36979	0.0000
Regime 2				
C	-0.236004	0.166934	-1.413759	0.1574
Common				
LOG(SIGMA)	-0.512333	0.066150	-7.745062	0.0000
Transition Matrix Parameters				
P11-C	3.441306	0.613213	5.611923	0.0000
P21-C	-2.211429	0.702318	-3.148760	0.0016
Mean dependent var	0.538792	S.D. dependent var	0.752727	
S.E. of regression	0.666025	Sum squared resid	74.96664	
Durbin-Watson stat	2.197989	Log likelihood	-174.1448	
Akaike info criterion	2.083079	Schwarz criterion	2.174576	
Hannan-Quinn criter.	2.120202			

Figure 28: Markov-switching model estimation for Spain

The results indicate that regime 1 has an estimated mean of 0.803 and is statistically significant at the 1% level. Regime 2, in turn, has an estimated mean of -0.236, although this parameter is not statistically significant. Nevertheless, the difference between the estimated means allows regime 1 to be associated with periods of economic expansion and regime 2 with episodes of economic contraction.

To identify the timing of the different states, the smoothed probabilities estimated by the model were analyzed; these are presented in Figure 29.

The smoothed probabilities show that regime 1 dominates for most of the sample. In particular, probabilities close to one are observed from the early 1980s to the early 1990s, again between approximately 1993 and 2008, and finally from 2013 to the end of the period analyzed. This indicates that the Spanish economy remained in an expansionary regime for most of the time.

By contrast, regime 2 is concentrated mainly in three episodes. The first corresponds to the late 1970s and early 1980s, the second is observed around the early 1990s recession, and the third coincides with the 2008–2009 international financial crisis and the subsequent European sovereign debt crisis, extending to approximately 2013. During these episodes, the probability of being in regime 2 increases considerably, reaching values close to one.

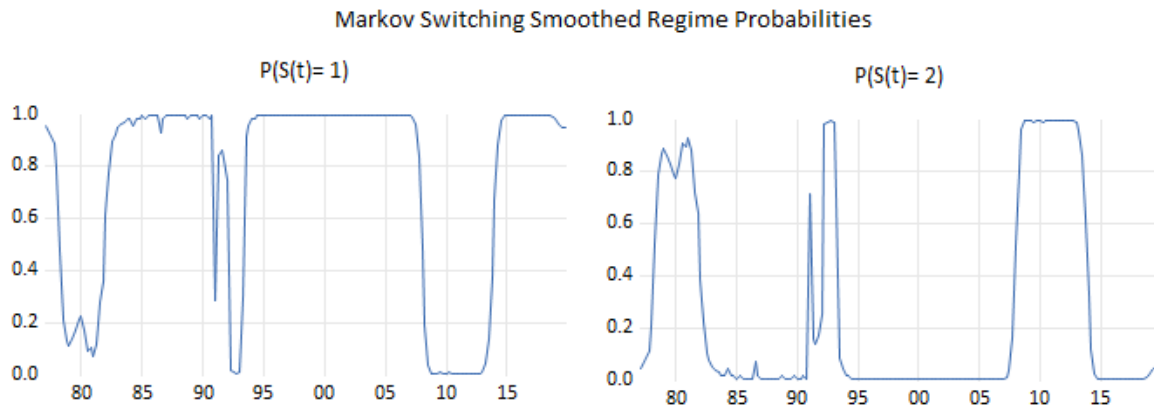


Figure 29: Smoothed regime probabilities for Spain

Overall, the model distinguishes a predominant regime of economic expansion and a second regime associated with periods of slowdown. Although the estimated mean for the latter is not statistically significant, its temporal identification coincides with some of the main crisis episodes observed in the Spanish economy over the sample period.

1.3.3 France

Following the same methodology applied above, a two-state Markov regime-switching model was estimated for France, allowing the mean to vary between regimes while the variance remains constant. The results obtained are presented in Figure 30.

Dependent Variable: FRANCE				
Method: Markov Switching Regression (BFGS / Marquardt steps)				
Date: 06/06/26 Time: 08:24				
Sample: 1977Q1 2019Q4				
Included observations: 172				
Number of states: 2				
Initial probabilities obtained from ergodic solution				
Standard errors & covariance computed using observed Hessian				
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=943195916)				
Convergence achieved after 5 iterations				
Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	-1.017476	0.252094	-4.036102	0.0001
Regime 2				
C	0.510682	0.035281	14.47485	0.0000
Common				
LOG(SIGMA)	-0.809809	0.056246	-14.39771	0.0000
Transition Matrix Parameters				
P11-C	0.856927	1.180467	0.725922	0.4679
P21-C	-4.896094	1.039931	-4.708094	0.0000
Mean dependent var	0.473555	S.D. dependent var	0.504789	
S.E. of regression	0.480599	Sum squared resid	39.03481	
Durbin-Watson stat	1.453553	Log likelihood	-113.1834	
Akaike info criterion	1.374225	Schwarz criterion	1.465722	
Hannan-Quinn criter.	1.411348			

Figure 30: Markov-switching model estimation for France

The results show that both regimes have statistically significant means at the 1% level and are clearly differentiated from each other. In particular, regime 1 has an estimated mean of -1.017, while regime 2 has a mean of 0.511. This difference allows two well-defined states to be identified in the dynamics of French economic growth.

From an economic standpoint, regime 1 can be interpreted as a recessionary state, given that it is associated with a negative average growth rate. Regime 2, in turn, can be interpreted as an expansionary regime, characterized by a positive average growth rate. The model also incorporates a single common variance for both states, in line with the proposed specification.

To identify the timing of the different regimes, the smoothed probabilities estimated by the model were analyzed; these are presented in Figure 31.

The smoothed probabilities show that regime 2 dominates for practically the entire sample, with probabilities close to one for most of the period analyzed. This indicates that the French economy remained predominantly in an expansionary regime, characterized by positive and relatively stable growth rates.

By contrast, regime 1 appears only in very specific episodes. The most clearly identified period corresponds to the 2008–2009 international financial crisis, during which the probability of being in the recessionary regime reaches values close to one. A shorter episode is also observed in the early 1980s, though considerably less pronounced than the one recorded during the global financial crisis.

The low frequency with which regime 1 is observed suggests high persistence of the expansionary regime and a low frequency of transitions between states. Consequently, the model identifies an economy that remains in a positive growth phase most of the time, interrupted only by exceptional episodes of economic deterioration associated with significant macroeconomic shocks.

In summary, the Markov regime-switching model clearly distinguishes between a predominant expansionary regime and a transitory recessionary regime, the latter being consistent with the main crisis episodes observed in the French economy over the period analyzed.

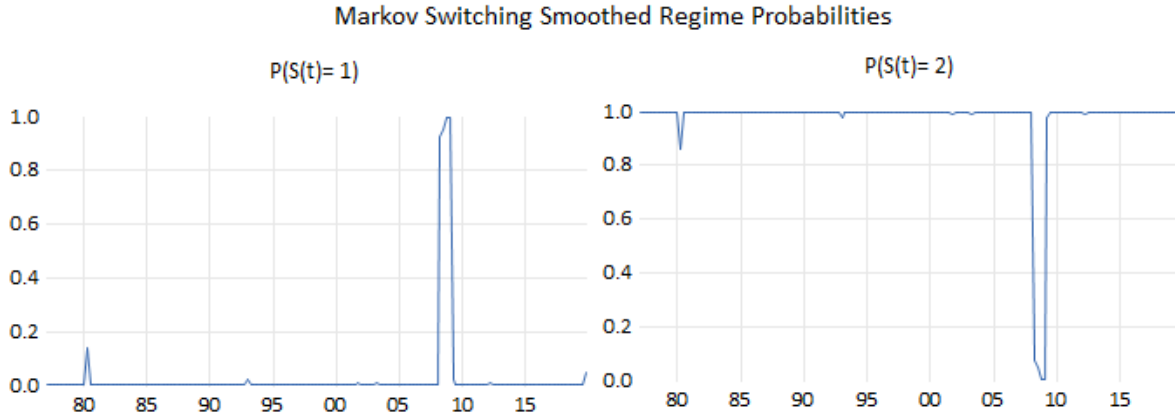


Figure 31: Smoothed regime probabilities for France

1.3.4 Finland

Finally, a two-state Markov regime-switching model was estimated for Finland under the same specification used for the other countries, allowing changes only in the mean of the series while keeping the variance constant. The results obtained are presented in Figure 32.

The results show that both regimes have statistically significant means that are clearly differentiated from each other. In particular, regime 1 has an estimated mean of 0.732, while regime 2 has a mean of -1.515. This difference evidences the existence of two well-defined states in the dynamics of Finnish economic growth.

From an economic standpoint, regime 1 can be interpreted as an expansionary state, characterized by a positive average growth rate. Regime 2, by contrast, is associated with a low-growth or recessionary state, given that it has a negative and statistically significant mean. The model also incorporates a single common variance for both regimes, in line with the proposed specification.

Dependent Variable: FINLAND				
Method: Markov Switching Regression (BFGS / Marquardt steps)				
Date: 06/06/26 Time: 08:26				
Sample: 1977Q1 2019Q4				
Included observations: 172				
Number of states: 2				
Initial probabilities obtained from ergodic solution				
Standard errors & covariance computed using observed Hessian				
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=943195916)				
Convergence achieved after 7 iterations				
Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	0.731654	0.096293	7.598231	0.0000
Regime 2				
C	-1.514643	0.515747	-2.936793	0.0033
Common				
LOG(SIGMA)	0.086223	0.058172	1.482205	0.1383
Transition Matrix Parameters				
P11-C	4.027726	0.752601	5.351741	0.0000
P21-C	-1.408119	0.838768	-1.678794	0.0932
Mean dependent var	0.543997	S.D. dependent var	1.258464	
S.E. of regression	1.186967	Sum squared resid	238.1024	
Durbin-Watson stat	2.081370	Log likelihood	-271.3914	
Akaike info criterion	3.213854	Schwarz criterion	3.305351	
Hannan-Quinn criter.	3.250976			

Figure 32: Markov-switching model estimation for Finland

To identify the timing of the different states, the smoothed probabilities estimated by the model were analyzed; these are presented in Figure 33.

The smoothed probabilities show that regime 1 dominates for most of the period analyzed, with probabilities close to one over most of the sample. This indicates that the Finnish economy remained predominantly in a positive-growth regime.

Regime 2, in turn, appears concentrated in specific episodes. The most clearly identified period corresponds to the recession of the early 1990s, when the probability of being in the recessionary regime reaches values close to one between approximately 1990 and 1992. Another relevant episode is observed during the 2008–2009 international financial crisis. Additionally, some isolated episodes of shorter duration can be identified in the late 1970s and at certain later periods, though with considerably lower intensity.

Consequently, the model identifies two clearly differentiated states for the Finnish economy. Unlike France, where the recessionary regime appears mainly during the international financial crisis, in Finland the most relevant episode corresponds to the deep recession of the early 1990s. This suggests that regime changes in the Finnish economy were associated not only with international shocks but also with structural disturbances that significantly affected its economic performance.

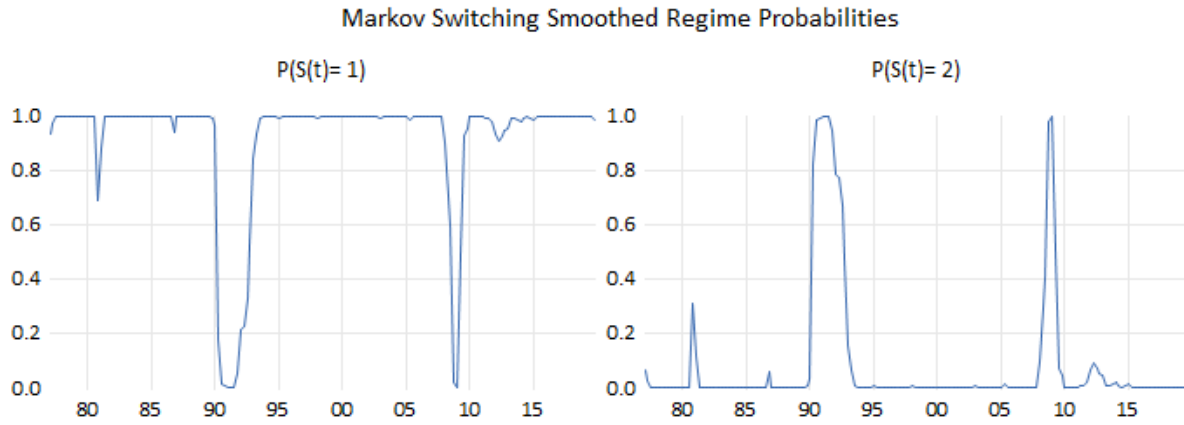


Figure 33: Smoothed regime probabilities for Finland

1.4 Business-cycle leadership

In order to analyze whether one of the economies tends to anticipate the other in expansion or recession phases, the smoothed probabilities obtained from the Markov regime-switching models estimated in the previous section were compared. In particular, the timing at which each economy transitions between regimes was examined, with the aim of identifying possible cyclical leadership relationships between the countries considered.

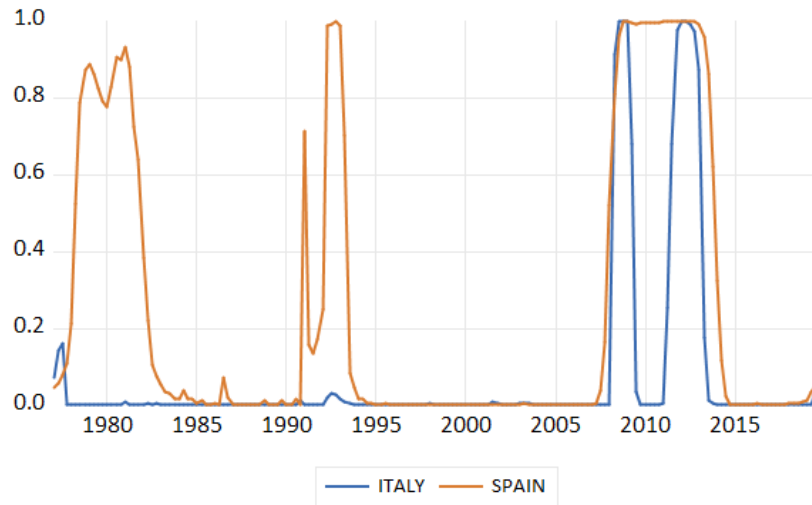


Figure 34: Smoothed probabilities of the recessionary regime: Italy and Spain

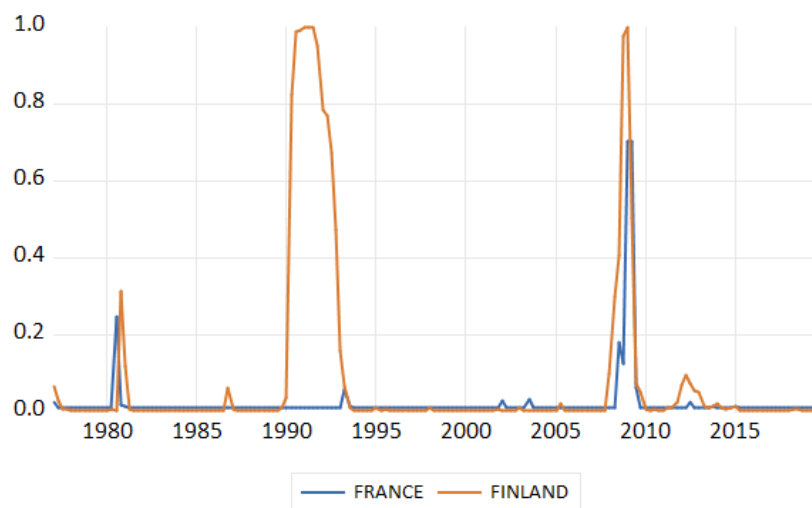


Figure 35: Smoothed probabilities of the recessionary regime: France and Finland

Figure 34 presents the smoothed probabilities associated with the recessionary regime for Italy and Spain. Both economies experience regime changes at very similar times, particularly during the 2008–2009 international financial crisis and the subsequent European sovereign debt crisis. While Spain appears to increase its probability of recession slightly earlier than Italy in some episodes, the time differences are small and not systematically maintained across the sample. There is therefore no conclusive evidence that either economy leads the other's cycle. Instead, the results suggest a high degree of synchronization between the business cycles of both countries.

Figure 35 presents the smoothed probabilities of the recessionary regime for France and Finland. Unlike the previous case, Finland experiences a deep recession in the early 1990s without France exhibiting a similar regime change, reflecting the existence of country-specific shocks that mainly affected the Finnish economy. During the 2008–2009 international financial crisis, however, both economies enter the recessionary regime, although Finland appears to do so with some lead time. In particular, the probability of recession rises first in Finland, while France registers the regime change some periods later. This behavior suggests some evidence of cyclical leadership by Finland during that episode. However, since this relationship is not observed systematically across the sample, the evidence of leadership is limited and should be interpreted with caution.

Overall, the results indicate that there is no strong and persistent cyclical leadership relationship between the countries analyzed. While Italy and Spain exhibit highly synchronized cycles, France and Finland show a more heterogeneous dynamic, with some evidence of Finland leading during the international financial crisis, though not consistently across the entire period analyzed.

1.5 Comparison of causality and cyclical leadership results

Comparing the results obtained for both country pairs reveals important differences in terms of both causality and business-cycle leadership.

For Italy and Spain, the Granger causality tests showed a bidirectional relationship between the GDP growth rates of both countries. Likewise, the regime-switching model analysis showed a high degree of cyclical synchronization, particularly during the 2008–2009 international financial crisis and the subsequent European sovereign debt crisis. While small time differences can be observed in some episodes, no conclusive evidence was found that either economy systematically leads the other’s cycle.

For the France-Finland pair, by contrast, evidence of unidirectional Granger causality running from France to Finland was obtained. However, the regime analysis showed lower synchronization between the two countries. In particular, Finland experienced a deep recession in the early 1990s that was not replicated by France, while during the 2008–2009 international financial crisis some evidence was found that the probability of being in the recessionary regime rose earlier in Finland than in France. However, this behavior was not maintained systematically across the sample, so the evidence of cyclical leadership is limited.

These differences may be related to the different degree of economic integration between the countries analyzed. Italy and Spain have strong trade, financial, and institutional ties, which favors greater synchronization of their business cycles and closer dynamic relationships. Finland, by contrast, has a more specialized productive structure and a relatively smaller economy, so certain shocks may affect it differently than they affect France.

The comparison of both approaches also shows that the conclusions derived from the Granger causality tests do not necessarily coincide with those obtained from the regime-leadership analysis. While Granger causality assesses whether the past values of one variable contain useful information for predicting another, cyclical leadership analysis focuses on the timing at which economies transition between expansion and recession phases. Consequently, one economy may provide relevant information for anticipating the evolution of another without necessarily leading its regime changes. Similarly, an economy may enter a recessionary phase earlier without this implying a strong Granger causality relationship.

It should also be noted that the regime-switching models estimated in this work allow only for changes in the mean, keeping the variance constant across states. Therefore, the identified regimes primarily reflect differences in the economies’ average growth, while the Granger causality tests use all the information contained in the temporal dynamics of the series. This methodological difference may also help explain why the results obtained through both approaches do not fully coincide.

Overall, the results suggest that Granger causality and regime leadership are complementary tools for analyzing macroeconomic interrelationships. While the former identifies predictive relationships between economies, the latter provides information on the synchronization and timing of their business cycles. Considered together, both methodologies offer a more complete view of the economic dynamics between the countries analyzed.

1.6 Analysis over the Great Moderation period

In order to evaluate whether the conclusions obtained above hold during the so-called Great Moderation period, the models for the Italy-Spain pair were re-estimated using only observations between 1984Q1 and 2007Q4. This period is characterized by lower macroeconomic volatility and greater growth stability in numerous developed economies.

1.6.1 ARMA modeling

As a first step in the analysis for the Great Moderation period (1984Q1–2007Q4), the univariate properties of Italy's and Spain's GDP growth rates were re-examined through their correlograms and the estimation of ARMA models.

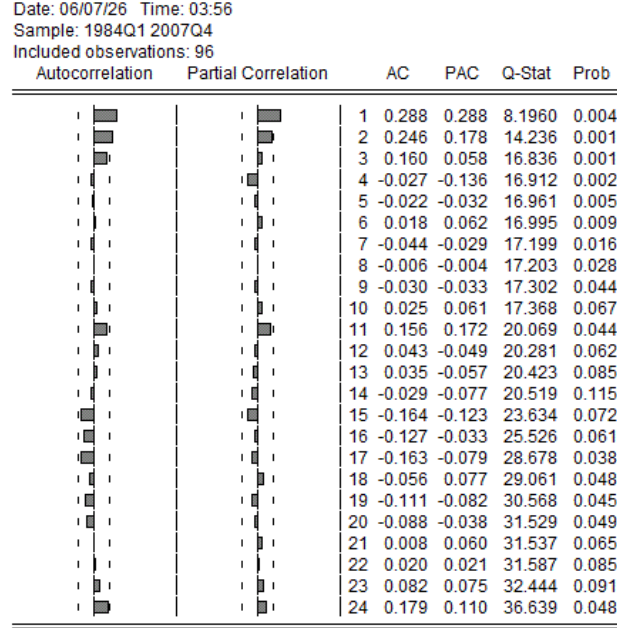


Figure 36: Correlogram for Italy

For Italy, the correlogram shown in Figure 36 continues to show relatively simple dynamics, characterized by a significant autocorrelation at the first lag and a rapid decline in subsequent correlations. Consequently, an AR(1) model was estimated, with results presented in Figure 37. The autoregressive coefficient is statistically significant, and the residual correlogram (Figure 37) shows no relevant remaining autocorrelation, indicating that the model adequately captures the dynamics of the series. Therefore, restricting the sample to the Great Moderation period does not substantially change the univariate specification obtained for the Italian economy, since an AR(1) model had also been selected for the full sample.

Date: 06/07/26 Time: 03:56
Sample: 1984Q1 2007Q4
Included observations: 96

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 0.288	0.288	8.1960	0.004
		2 0.246	0.178	14.236	0.001
		3 0.160	0.058	16.836	0.001
		4 -0.027	-0.136	16.912	0.002
		5 -0.022	-0.032	16.961	0.005
		6 0.018	0.062	16.995	0.009
		7 -0.044	-0.029	17.199	0.016
		8 -0.006	-0.004	17.203	0.028
		9 -0.030	-0.033	17.302	0.044
		10 0.025	0.061	17.368	0.067
		11 0.156	0.172	20.069	0.044
		12 0.043	-0.049	20.281	0.062
		13 0.035	-0.057	20.423	0.085
		14 -0.029	-0.077	20.519	0.115
		15 -0.164	-0.123	23.634	0.072
		16 -0.127	-0.033	25.526	0.061
		17 -0.163	-0.079	28.678	0.038
		18 -0.056	0.077	29.061	0.048
		19 -0.111	-0.082	30.568	0.045
		20 -0.088	-0.038	31.529	0.049
		21 0.008	0.060	31.537	0.065
		22 0.020	0.021	31.587	0.085
		23 0.082	0.075	32.444	0.091
		24 0.179	0.110	36.639	0.048

(a) Correlogram of the AR(1) residuals

Dependent Variable: ITALY
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/07/26 Time: 03:58
Sample: 1984Q1 2007Q4
Included observations: 96
Convergence achieved after 5 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.446878	0.074670	5.984741	0.0000
AR(1)	0.293968	0.097936	3.001639	0.0034
SIGMASQ	0.262727	0.044551	5.897182	0.0000
R-squared	0.085471	Mean dependent var	0.452515	
Adjusted R-squared	0.065804	S.D. dependent var	0.538799	
S.E. of regression	0.520770	Akaike info criterion	1.564677	
Sum squared resid	25.22175	Schwarz criterion	1.644813	
Log likelihood	-72.10449	Hannan-Quinn criter.	1.597069	
F-statistic	4.345849	Durbin-Watson stat	2.083010	
Prob(F-statistic)	0.015692			
Inverted AR Roots	.29			

(b) Estimated AR(1) equation

Figure 37: AR(1) model for Italy during the Great Moderation

For Spain, the results differ from those obtained for the full sample. The correlogram of the series, presented in Figure 38, suggests a more persistent temporal structure than that observed for Italy, leading to the estimation of an AR(3) model, with results shown in Figure 39. All three autoregressive coefficients are statistically significant, and the inverse roots remain within the unit circle, guaranteeing the stationarity of the process. The residual correlogram (Figure 39) also shows that the selected specification captures most of the temporal dependence in the series, although a few isolated signs of autocorrelation remain.

Comparison with the results obtained for the full sample suggests that, during the Great Moderation, the dynamics of Spanish economic growth became relatively more stable and less complex. In particular, while an ARMA(3,1) model had been selected for the full sample, an AR(3) model is sufficient to adequately describe the behavior of the series over 1984Q1–2007Q4. This result is consistent with the lower macroeconomic volatility characteristic of the Great Moderation. For Italy, by contrast, the optimal specification remains unchanged between both samples, suggesting greater stability in the temporal structure of its growth rate.

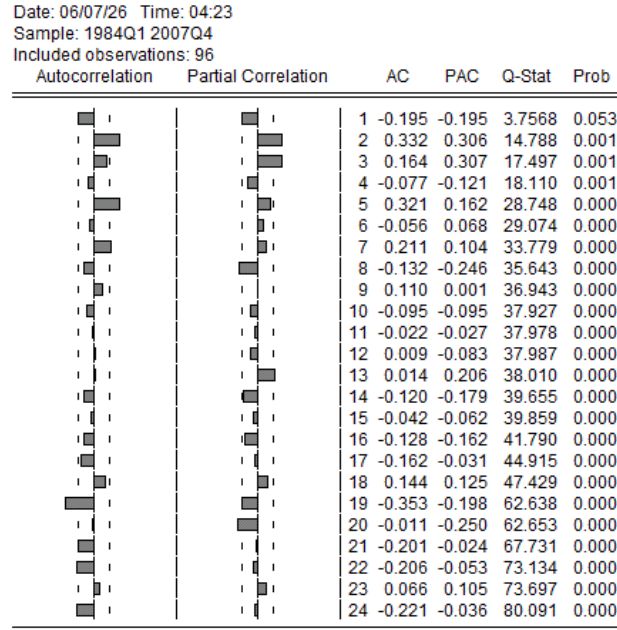
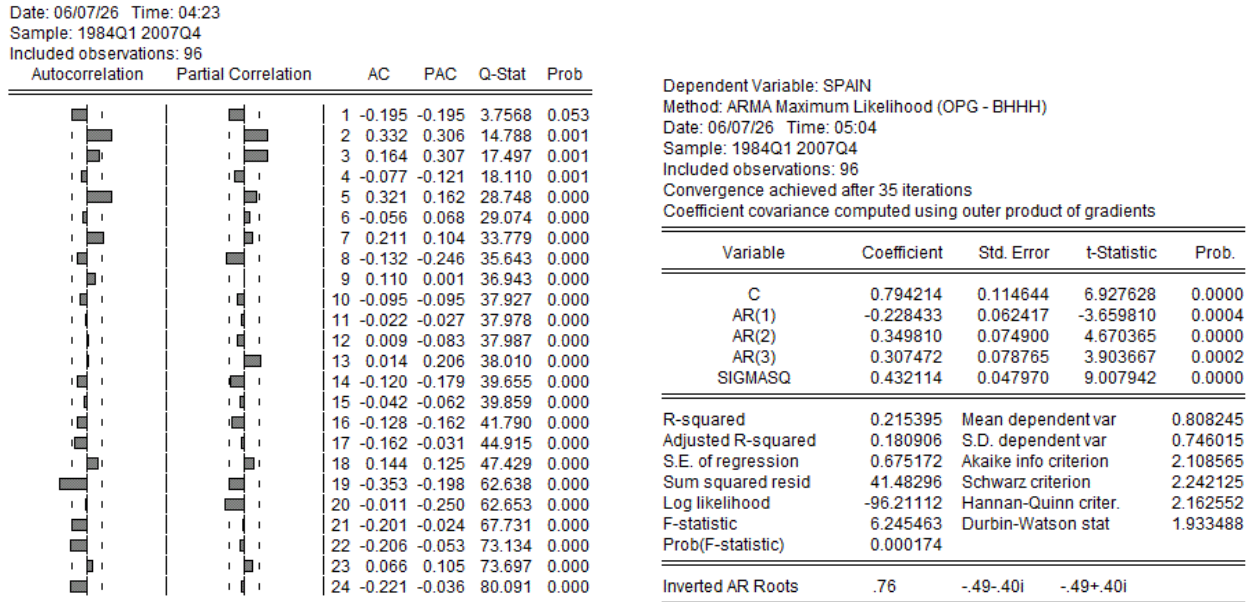


Figure 38: Correlogram for Spain



(a) Correlogram of the AR(3) residuals

(b) Estimated AR(3) equation

Figure 39: AR(3) model for Spain during the Great Moderation

1.6.2 VAR model

To analyze the dynamic interrelationships between the GDP growth rates of Italy and Spain during the Great Moderation, a Vector Autoregression (VAR) model was estimated using the restricted sample between 1984Q1 and 2007Q4.

As a preliminary step, the optimal number of lags was determined using the Akaike (AIC), Schwarz (SC), Hannan-Quinn (HQ), final prediction error (FPE), and sequential likelihood ratio (LR) test criteria. The results are presented in Figure 40.

VAR Lag Order Selection Criteria
Endogenous variables: ITALY SPAIN
Exogenous variables: C
Date: 06/07/26 Time: 05:08
Sample: 1984Q1 2007Q4
Included observations: 96

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-183.5784	NA	0.163730	3.866216	3.919640	3.887811
1	-169.8127	26.67104	0.133594	3.662764	3.823036*	3.727549
2	-162.8377	13.22338*	0.125584	3.600786	3.867905	3.708760*
3	-158.7590	7.562698	0.125420*	3.599145*	3.973113	3.750309
4	-156.9186	3.335624	0.131268	3.644138	4.124953	3.838491
5	-154.8603	3.644936	0.136813	3.684590	4.272253	3.922133
6	-151.2864	6.179909	0.138216	3.693466	4.387977	3.974199

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Figure 40: Lag-order selection criteria for the Italy–Spain VAR

The LR, FPE, AIC, and HQ criteria select a specification with two lags, while the Schwarz criterion favors a more parsimonious model with a single lag. Since most criteria agree on selecting two lags, and considering that this specification adequately captures the dynamic interactions between the two economies, a VAR(2) model was estimated.

It is worth noting that this choice differs from that obtained for the full sample, where a VAR(3) had been selected. This suggests that, during the Great Moderation, the joint dynamics of the Italian and Spanish economies can be described by a less persistent temporal structure, consistent with the greater macroeconomic stability characteristic of this period.

The VAR(2) estimation results are presented in Figure 41. In the equation for Italy, the first lag of Spanish growth is statistically significant and positive, while the lags of Italian growth are not individually significant. In the equation for Spain, the second lags of both Italian and Spanish growth are statistically significant and positive. These results suggest the existence of dynamic interactions between the two economies, although with a different structure than that observed in the full sample.

Subsequently, the stability of the system was verified by analyzing the roots of the characteristic polynomial. As shown in Figure 42(a), none of the roots has a modulus greater than one. Likewise, the graphical representation of the inverse roots shows that all of them lie within the unit circle, as seen in Figure 42(b). The stability condition of the system is therefore verified, allowing valid inferences to be drawn from the estimated model.

Once the stability of the VAR(2) was confirmed, the existence of Granger causality relationships between the two economies was analyzed. The results are presented in Figure 43. First, the null hypothesis that the lags of Spain's GDP growth rate do not Granger-cause Italy's GDP growth rate is tested. The p-value associated with this test is 0.0016, so the null hypothesis is rejected at the 1% significance level. There is therefore evidence that information contained in the past values of Spanish economic growth helps explain the future

evolution of Italian growth.

Similarly, the null hypothesis that the lags of Italy's GDP growth rate do not Granger-cause Spain's GDP growth rate is evaluated. The resulting p-value is 0.0128, rejecting the null hypothesis at the 5% significance level. There is therefore evidence that past values of Italian growth provide relevant information for predicting future Spanish growth.

Vector Autoregression Estimates
Date: 06/07/26 Time: 05:09
Sample: 1984Q1 2007Q4
Included observations: 96
Standard errors in () & t-statistics in []

	ITALY	SPAIN
ITALY(-1)	0.137012 (0.09875) [1.38750]	0.159525 (0.13513) [1.18049]
ITALY(-2)	0.058010 (0.09412) [0.61635]	0.296428 (0.12880) [2.30144]
SPAIN(-1)	0.248435 (0.07193) [3.45397]	-0.208494 (0.09843) [-2.11815]
SPAIN(-2)	0.122804 (0.07291) [1.68424]	0.247083 (0.09978) [2.47623]
C	0.058780 (0.10429) [0.56365]	0.554932 (0.14271) [3.88839]
R-squared	0.186464	0.205261
Adj. R-squared	0.150704	0.170328
Sum sq. resids	22.43647	42.01871
S.E. equation	0.496543	0.679518
F-statistic	5.214336	5.875762
Log likelihood	-66.44241	-96.55891
Akaike AIC	1.488383	2.115811
Schwarz SC	1.621943	2.249370
Mean dependent	0.452515	0.808245
S.D. dependent	0.538799	0.746015
Determinant resid covariance (dof adj.)		0.113458
Determinant resid covariance		0.101947
Log likelihood		-162.8377
Akaike information criterion		3.600786
Schwarz criterion		3.867905
Number of coefficients		10

Figure 41: VAR(2) for Italy and Spain during the Great Moderation period

As in the full sample, the results indicate bidirectional Granger causality between the two economies. However, the reduction in the optimal number of lags suggests that these dynamic relationships operate through a simpler temporal structure during the Great Moderation.

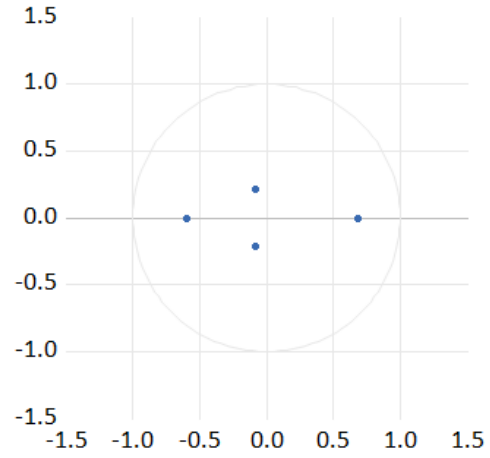
Roots of Characteristic Polynomial
 Endogenous variables: ITALY SPAIN
 Exogenous variables: C
 Lag specification: 1 2
 Date: 06/07/26 Time: 05:12

Root	Modulus
0.688737	0.688737
-0.598552	0.598552
-0.080834 - 0.216794i	0.231374
-0.080834 + 0.216794i	0.231374

No root lies outside the unit circle.
 VAR satisfies the stability condition.

(a) Eigenvalues of the VAR(2)

Inverse Roots of AR Characteristic Polynomial



(b) Plot of the VAR(2) eigenvalues

Figure 42: Stability analysis of the VAR(2) system during the Great Moderation

VAR Granger Causality/Block Exogeneity Wald Tests
 Date: 06/07/26 Time: 05:11
 Sample: 1984Q1 2007Q4
 Included observations: 96

Dependent variable: ITALY

Excluded	Chi-sq	df	Prob.
SPAIN	12.82704	2	0.0016
All	12.82704	2	0.0016

Dependent variable: SPAIN

Excluded	Chi-sq	df	Prob.
ITALY	8.710068	2	0.0128
All	8.710068	2	0.0128

Figure 43: Granger causality test for Italy – Spain

In order to analyze the dynamic transmission of disturbances between the Italian and Spanish economies, impulse-response functions (IRFs) were estimated based on the VAR(2) model. As in the full sample, results are reported for both possible orderings of the Cholesky decomposition, Italy-Spain and Spain-Italy, in order to evaluate the robustness of the results to the ordering used.

Figures 44 and 45 present the impulse-response functions obtained for both orderings. In general terms, the responses follow the same pattern observed for the full sample. Own innovations generate positive and

immediate responses that gradually dissipate until converging to zero, while cross responses show evidence of shock transmission between the two economies.

However, compared with the full sample, the responses show lower temporal persistence and converge more quickly to equilibrium. This result is consistent with the lower volatility and greater macroeconomic stability that characterized the Great Moderation. Likewise, the main conclusions remain unchanged when the ordering of the variables is altered, suggesting that the results are relatively robust to shock identification.

Overall, the results obtained for the 1984Q1–2007Q4 period show that the Italian and Spanish economies continued to exhibit strong dynamic interdependence, reflected both in bidirectional Granger causality and in the transmission of shocks between the two economies. However, the system's dynamics are less persistent than in the full sample, consistent with the Great Moderation phenomenon documented for developed economies.

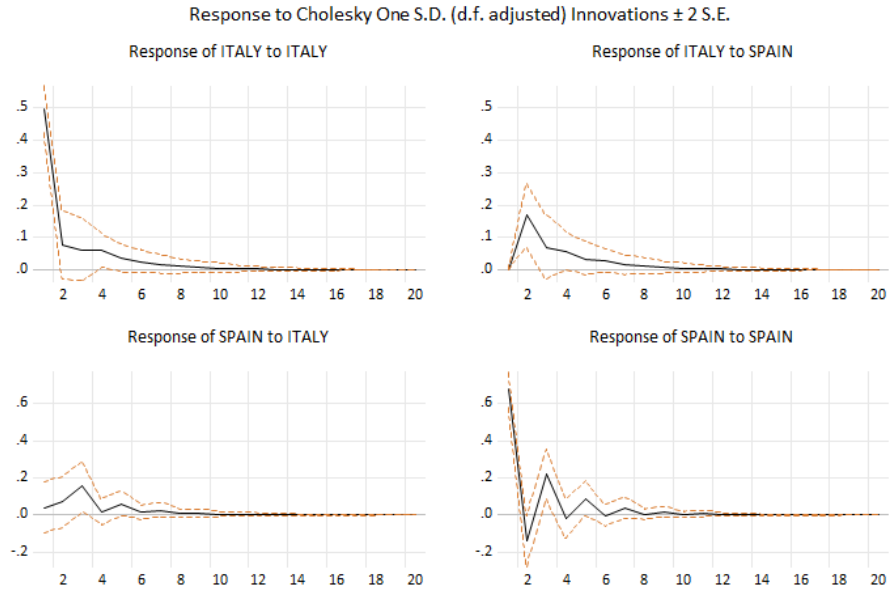


Figure 44: Impulse-response functions with Italy-Spain ordering

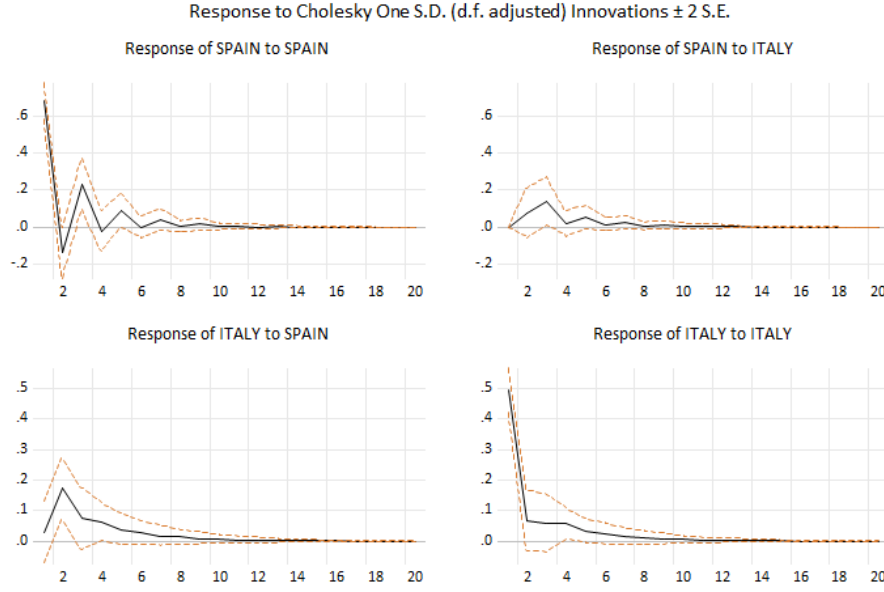


Figure 45: Impulse-response functions with Spain-Italy ordering

1.6.3 Markov regime-switching models

To identify possible regime changes in the GDP growth rates of Italy and Spain during the Great Moderation, two-state Markov regime-switching models were estimated, allowing only the mean to differ between regimes while the variance remains constant. The estimation results are presented in Figures 46 and 48, while the smoothed probabilities associated with each regime are shown in Figures 47 and 49.

For Italy (Figure 46), the results indicate that regime 1 has an estimated mean of 0.809, significant at the 1% level, while the estimated mean for regime 2 is 0.040 and is not statistically significant. This implies that there is not enough statistical evidence to conclude that the average growth rate associated with the second regime differs from zero. Consequently, regime 1 can be interpreted as a state of economic expansion, while regime 2 is characterized by a considerably lower average growth rate than that observed in regime 1. However, since that mean is not statistically significant, the differentiation between the two states is less pronounced than in the full sample. The model also incorporates a single common variance for both states, in line with the specification used.

The smoothed probabilities presented in Figure 47 show relatively frequent alternation between both regimes throughout the sample. Unlike the results obtained for the full sample, where recessionary episodes clearly associated with the 2008–2009 international financial crisis and the European debt crisis were identified, the Great Moderation subsample does not show extended periods of remaining in a recessionary regime. This result is consistent with the greater macroeconomic stability characteristic of the period analyzed.

Dependent Variable: ITALY
Method: Markov Switching Regression (BFGS / Marquardt steps)
Date: 06/07/26 Time: 05:57
Sample: 1984Q1 2007Q4
Included observations: 96
Number of states: 2
Initial probabilities obtained from ergodic solution
Standard errors & covariance computed using observed Hessian
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=2083842769)
Convergence achieved after 3 iterations

Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	0.808915	0.077271	10.46861	0.0000
Regime 2				
C	0.040107	0.094337	0.425146	0.6707
Common				
LOG(SIGMA)	-0.981989	0.103991	-9.443000	0.0000
Transition Matrix Parameters				
P11-C	1.478172	0.504024	2.932743	0.0034
P21-C	-1.377573	0.558788	-2.465289	0.0137
Mean dependent var	0.452515	S.D. dependent var	0.538799	
S.E. of regression	0.508057	Sum squared resid	24.00530	
Durbin-Watson stat	2.026855	Log likelihood	-68.49366	
Akaike info criterion	1.531118	Schwarz criterion	1.664678	
Hannan-Quinn criter.	1.585105			

Figure 46: Markov-switching model estimation for Italy

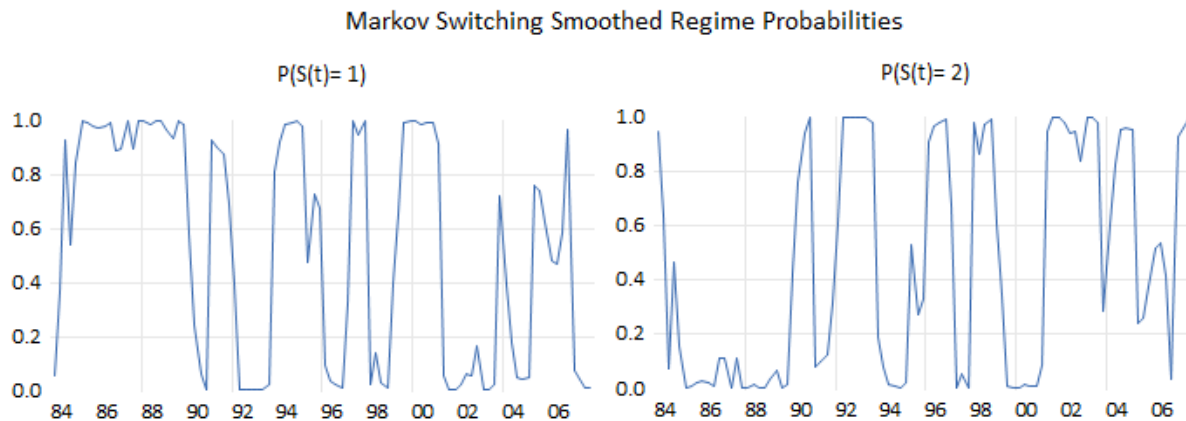


Figure 47: Smoothed regime probabilities for Italy

For Spain, the results show a more pronounced differentiation between the regimes (Figure 48). Regime 1 has an estimated mean of -0.826, significant at the 1% level, while regime 2 exhibits a positive mean of 0.917, also significant at the 1% level. Consequently, regime 1 can be interpreted as a recessionary state, while regime 2 is associated with periods of expansion.

The smoothed probabilities presented in Figure 49 indicate that the expansionary regime dominates for most of the period analyzed. Episodes associated with the contractionary regime are concentrated mainly between 1990 and 1993, coinciding with the economic slowdown that affected several European economies in the early 1990s. Compared with the full sample, the extended recessionary episodes associated with the international financial crisis and the European sovereign debt crisis disappear, reflecting the relatively stable nature of the Great Moderation period.

Dependent Variable: SPAIN
Method: Markov Switching Regression (BFGS / Marquardt steps)
Date: 06/07/26 Time: 06:05
Sample: 1984Q1 2007Q4
Included observations: 96
Number of states: 2
Initial probabilities obtained from ergodic solution
Standard errors & covariance computed using observed Hessian
Random search: 100 starting values with 10 iterations using 1 standard deviation (rng=kn, seed=2083842769)
Convergence achieved after 4 iterations

Variable	Coefficient	Std. Error	z-Statistic	Prob.
Regime 1				
C	-0.826484	0.294471	-2.806671	0.0050
Regime 2				
C	0.917052	0.068240	13.43865	0.0000
Common				
LOG(SIGMA)	-0.493267	0.079529	-6.202390	0.0000
Transition Matrix Parameters				
P11-C	0.186474	0.977419	0.190782	0.8487
P21-C	-3.492902	0.766866	-4.554774	0.0000
Mean dependent var	0.808245	S.D. dependent var	0.746015	
S.E. of regression	0.744493	Sum squared resid	51.54713	
Durbin-Watson stat	2.778383	Log likelihood	-101.8183	
Akaike info criterion	2.225380	Schwarz criterion	2.358940	
Hannan-Quinn criter.	2.279367			

Figure 48: Markov-switching model estimation for Spain

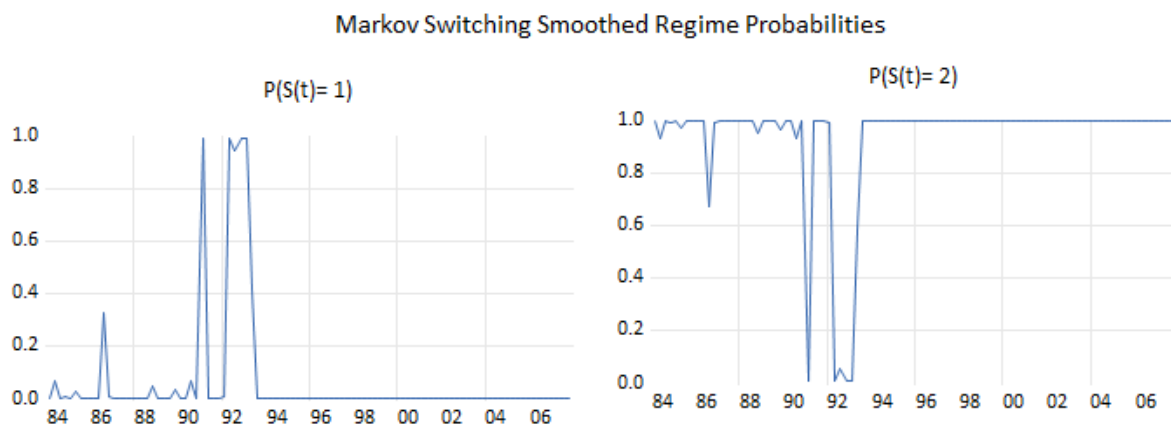


Figure 49: Smoothed regime probabilities for Spain

Finally, to analyze the possible existence of cyclical leadership between the two economies, the smoothed probabilities associated with the expansionary regimes identified for each country were compared. The results are presented in Figure 50.

As can be seen, the Spanish economy remains for most of the period with a probability close to one of being in the expansionary regime, with only one significant interruption around 1991–1993. Italy, in turn, shows greater alternation between states, with more frequent episodes of reduced probability of remaining in the expansionary regime.

Despite these differences, no clear and persistent temporal pattern is observed that would allow the conclusion that one economy systematically anticipates the other in regime changes. In particular, episodes of lower probability of expansion do not show a stable sequence in which one economy enters a slowdown phase first and the other follows with a lag. Consequently, as in the full sample, no conclusive evidence of cyclical leadership between Italy and Spain is found.

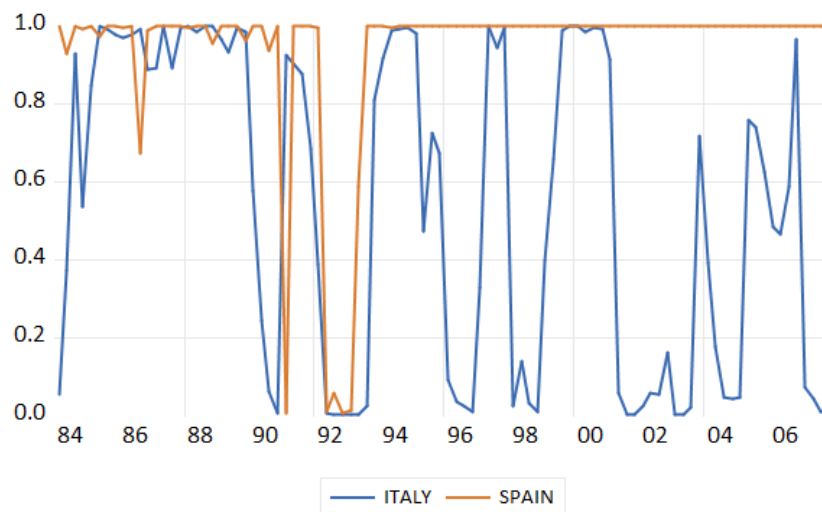


Figure 50: Smoothed probabilities of the expansionary regime: Italy and Spain

Overall, the results obtained for the 1984Q1–2007Q4 period differ from those observed for the full sample. In particular, the regimes appear less differentiated, especially for Italy, where the identification of a clearly recessionary state disappears. There is also lower persistence of episodes associated with low growth and a greater predominance of states compatible with economic expansions. This evidence is consistent with the literature on the Great Moderation, which documents a significant reduction in macroeconomic volatility and greater growth stability in developed economies during this period.

Likewise, comparing the results obtained through the Granger causality tests and the regime-leadership analysis reveals a situation similar to that found for the full sample. While Granger causality continues to indicate a bidirectional relationship between the GDP growth rates of Italy and Spain, the analysis of smoothed probabilities does not identify conclusive evidence of cyclical leadership between the two economies. This can be attributed to the fact that both approaches capture different aspects of economic dynamics: while

Granger causality assesses predictive relationships between the series, regime analysis focuses on the moments of transition between expansion and slowdown states. Additionally, the lower volatility characteristic of the Great Moderation generates less differentiated regimes, making it harder to identify clear leadership patterns.

2 U.S. and U.K. Interest Rates: Term Structure and Dynamic Analysis

2.1 Stationarity of U.S. and U.K. interest rates

In order to analyze the stochastic properties of the U.S. and U.K. interest rate term structures, the stationarity of rates at different maturities was studied, as well as that of the rate differentials between the two countries. In particular, very short-term rates (1, 3, 6, and 12 months) and long-term rates (2, 5, and 10 years) were considered over the period 2009M01–2025M12, in order to evaluate possible differences in their dynamic behavior. The stationarity of the series was analyzed through graphical inspection and via the Augmented Dickey-Fuller (ADF) test, whose null hypothesis establishes the presence of a unit root.

2.1.1 U.S. interest rates

Figure 51 presents the evolution of U.S. interest rates for the maturities considered. In general, all the series exhibit high temporal persistence and show significant level shifts throughout the sample. In particular, the very short-term rates (1, 3, 6, and 12 months) remain close to zero for several years, rise gradually toward the end of the 2010s, decline again during the COVID-19 pandemic, and rise sharply starting in 2022. The longer-maturity rates (2, 5, and 10 years) show a similar evolution, though with more pronounced fluctuations throughout the period analyzed.

To formally evaluate the stationarity of the series, the Augmented Dickey-Fuller (ADF) test was applied. This test contrasts the null hypothesis that the series has a unit root against the alternative hypothesis of stationarity. Consequently, rejecting the null hypothesis constitutes evidence in favor of the stationarity of the series, while failing to reject it suggests non-stationary behavior.

The test specification was selected based on visual inspection of the series. Although the rates show significant level shifts throughout the sample, no clearly defined linear deterministic trend is observed. For this reason, the ADF tests were estimated including a constant but excluding a trend term. This evaluates the presence of a unit root around a constant mean.

The results of the ADF tests for the different U.S. interest rates are summarized in Table 1, while the complete EViews outputs are presented in Appendix A.1. As can be observed, the ADF statistics obtained for all series are smaller in absolute value than the critical values, and the associated p-values are considerably higher than conventional significance levels. Consequently, the null hypothesis of a unit root is not rejected for any of the rates analyzed.

These results suggest that U.S. interest rates are non-stationary in levels, regardless of the maturity considered.

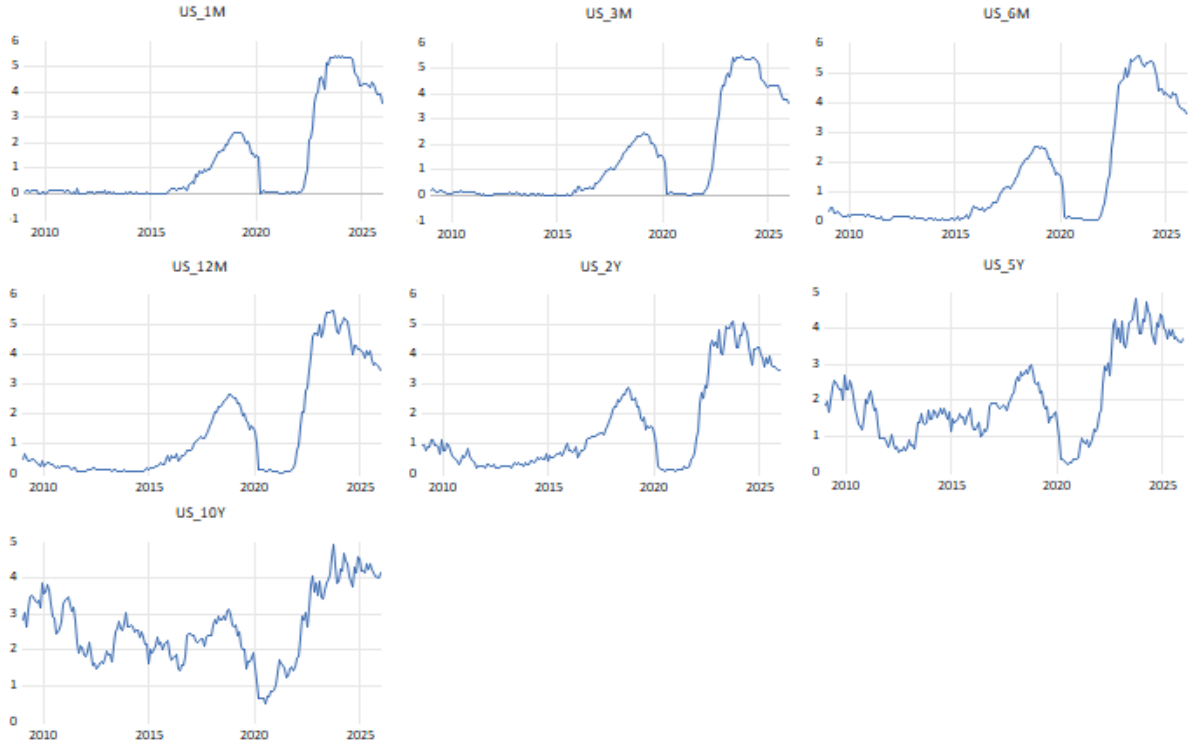


Figure 51: U.S. interest rates

Table 1: ADF statistics and p-values for U.S. interest rates

Series	ADF statistic	p-value
US.1M	-1.4388	0.5626
US.3M	-1.6493	0.4555
US.6M	-1.2133	0.6688
US.12M	-1.1291	0.7043
US.2Y	-0.6508	0.8551
US.5Y	-1.2159	0.6677
US.10Y	-1.4536	0.5552

2.1.2 U.K. interest rates

Figure 52 presents the evolution of U.K. interest rates for the maturities considered. As with the United States, the series exhibit high temporal persistence and show significant level shifts throughout the period analyzed. Very short-term rates remain relatively stable and at low levels for several years, then rise sharply starting in 2022. Long-term rates, in turn, show a more fluctuating evolution, though they also reflect the significant increase observed in the final years of the sample.

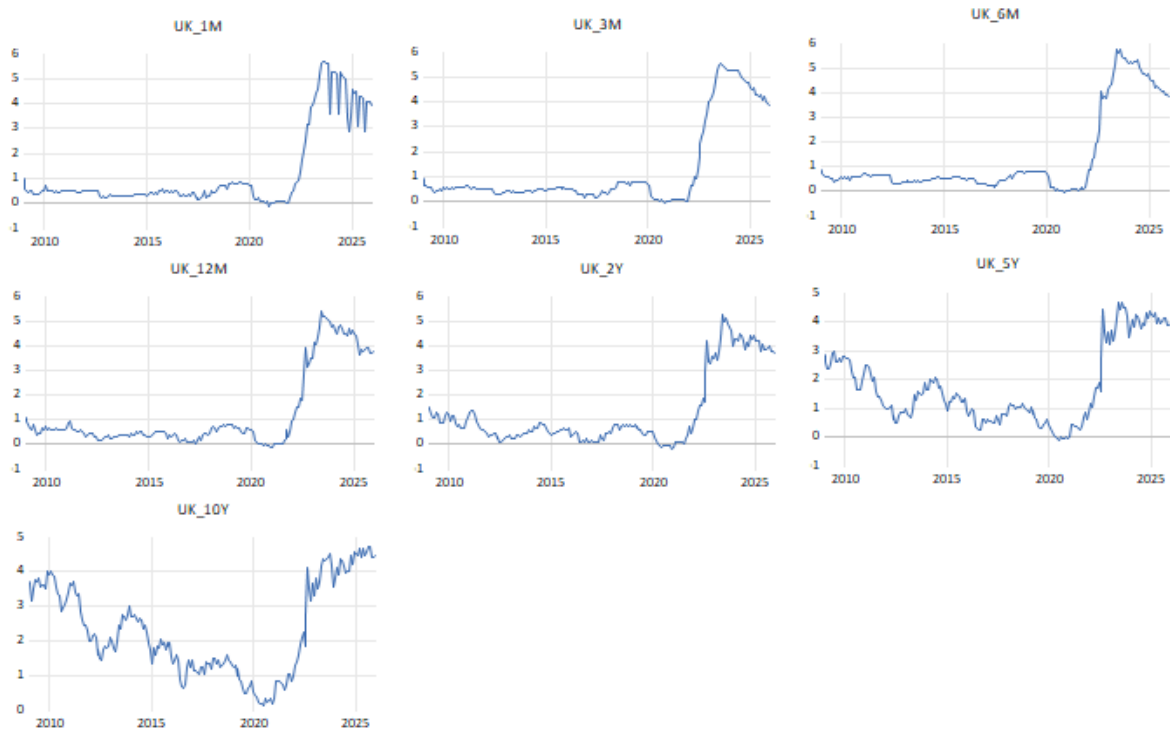


Figure 52: U.K. interest rates

Table 2: ADF statistics and p-values for U.K. interest rates

Series	ADF statistic	p-value
UK_1M	-1.1916	0.6782
UK_3M	-0.9218	0.7798
UK_6M	-0.4552	0.8958
UK_12M	-0.1415	0.9421
UK_2Y	-0.4491	0.8969
UK_5Y	-0.9609	0.7669
UK_10Y	-0.9448	0.7723

The stationarity of the series was evaluated using the Augmented Dickey-Fuller (ADF) test, using the same specification adopted for the U.S. rates. In particular, the tests were estimated including a constant and excluding a trend term, since visual inspection of the series does not show a clearly defined linear deterministic trend.

Table 2 shows the ADF statistics and p-values for the different U.K. interest rates, while the complete EViews outputs are presented in Appendix A.2. As shown in Table 2, the statistics obtained are considerably smaller in absolute value than the critical values in all cases, and the associated p-values greatly exceed conventional significance levels. Consequently, the null hypothesis of a unit root is not rejected for any of the rates analyzed.

Therefore, the empirical evidence suggests that U.K. interest rates are non-stationary in levels for all matu-

rities considered. As with the United States, the null hypothesis of a unit root is not rejected for any of the maturities analyzed.

2.1.3 Interest rate differentials

Figure 53 presents the evolution of the interest rate differentials between the United States and the United Kingdom for the maturities considered. In general terms, the differentials show different dynamics than those observed for the individual rates. While they also exhibit significant level shifts throughout the sample, their fluctuations appear to be bounded within relatively narrower ranges, particularly at longer maturities.

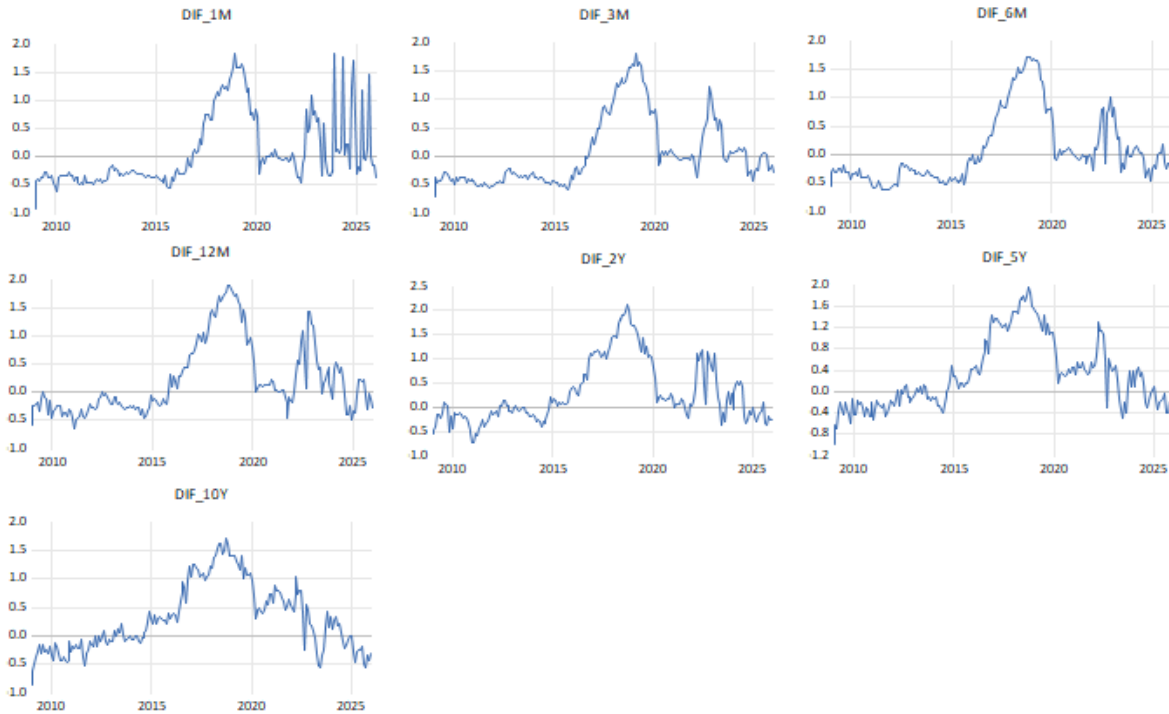


Figure 53: Interest rate differentials

The stationarity of the differentials was evaluated using the Augmented Dickey-Fuller (ADF) test, using the same specification adopted for the individual rates. Consequently, the tests were estimated including a constant and excluding a trend term, since no clearly defined linear deterministic trend is observed in the series.

The ADF statistics and p-values for the rate differentials are summarized in Table 3, while the complete EViews outputs are presented in Appendix A.3. The statistics obtained do not allow rejection of the null hypothesis of a unit root for any of the maturities considered, since the p-values are above conventional significance levels.

Therefore, the empirical evidence suggests that the rate differentials between the United States and the United Kingdom are also non-stationary in levels. However, it is observed that the differentials corresponding to longer maturities have relatively more negative ADF statistics and lower p-values than those obtained for

short-term differentials. Although this evidence is not sufficient to conclude stationarity, it could indicate a greater tendency toward reversion to equilibrium levels in the longer segments of the yield curve.

Table 3: ADF statistics and p-values for the interest rate differentials

Series	ADF statistic	p-value
DIF_1M	-1.7473	0.4059
DIF_3M	-1.7165	0.4214
DIF_6M	-1.9373	0.3147
DIF_12M	-2.2414	0.1925
DIF_2Y	-2.2588	0.1865
DIF_5Y	-2.5314	0.1096
DIF_10Y	-2.3684	0.1521

2.1.4 Comparison of results

Comparing the results obtained for the United States, the United Kingdom, and the rate differentials makes it possible to draw some relevant conclusions. First, the ADF tests do not allow rejection of the null hypothesis of a unit root for any of the interest rates considered, regardless of country or maturity. Both very short-term and long-term rates therefore show a high degree of persistence and behavior compatible with non-stationarity in levels.

However, graphical analysis suggests important differences between the different segments of the term structure of rates. Very short-term rates show more abrupt level shifts and prolonged periods at extremely low values, followed by significant increases over relatively short periods. This behavior is consistent with the influence of monetary policy decisions implemented by central banks, which more directly affect shorter-maturity rates.

Long-term rates, by contrast, exhibit more gradual trajectories and more persistent fluctuations over time. A possible explanation is that these rates reflect not only current monetary conditions but also expectations regarding the future evolution of macroeconomic variables such as inflation, economic growth, and long-term financial conditions. Consequently, their movements tend to be smoother and distributed over longer time horizons.

The interest rate differentials between the United States and the United Kingdom are also non-stationary for all maturities considered. However, the long-term differentials show relatively more negative ADF statistics and lower p-values than those observed for short-term differentials. Although this evidence is not sufficient to conclude stationarity, it could suggest a greater tendency toward reversion to equilibrium levels in the long segments of the yield curve.

Overall, the results indicate that the interest rates of both countries share a high degree of temporal persistence, regardless of maturity. While the differences between short- and long-term rates do not translate into different conclusions regarding stationarity, they do manifest in the dynamics observed in the series and in the behavior of the differentials, suggesting that different economic factors may influence each segment of the term structure of rates.

2.2 Cointegration analysis between interest rates

Since the previous section found evidence that interest rates at different maturities are non-stationary in levels and compatible with processes integrated of order one, $I(1)$, it is relevant to analyze whether a long-run equilibrium relationship exists between them. For this purpose, the Johansen cointegration methodology was applied, which allows evaluating the existence of stationary linear combinations among a set of non-stationary variables.

The presence of cointegration implies that, although the series may deviate temporarily from one another, there is a long-run equilibrium relationship that limits such divergences. In the context of the term structure of interest rates, the existence of cointegration relationships suggests that rates corresponding to different maturities share common stochastic trends and evolve jointly in the long run.

The Johansen test makes it possible to determine the number of cointegrating vectors present in the system using two statistics: the Trace statistic and the Maximum Eigenvalue statistic. This work primarily uses the results of the Trace statistic, which sequentially tests the null hypothesis that there are at most (r) cointegrating vectors against the alternative that there are more than (r) . The decision is based on comparing the calculated statistic with its critical value at the 5% significance level. When the statistic exceeds the critical value, the null hypothesis is rejected and it is concluded that a larger number of cointegration relationships exists.

To facilitate the presentation of results, a summary of the Trace statistics, critical values, and decisions associated with each hypothesis is presented for each country and for the rate differential. The complete EViews outputs of the Johansen tests are included in Appendix B, where the results of the Maximum Eigenvalue statistic can also be found.

2.2.1 United States

The results of the Johansen cointegration test for U.S. interest rates are summarized in Table 4. The Trace statistic successively rejects the null hypotheses of no cointegration ($r=0$), of at most one cointegrating vector ($r \leq 1$), and of at most two cointegrating vectors ($r \leq 2$). However, the null hypothesis that there are at most three cointegrating vectors ($r \leq 3$) is not rejected. Since this is the first hypothesis for which the null is not rejected, it is concluded that the system has three cointegrating vectors.

This result indicates that, although interest rates at different maturities are individually non-stationary, there are linear combinations among them that are indeed stationary. In other words, the rates share common stochastic trends and maintain long-run equilibrium relationships that limit divergence between the different segments of the yield curve.

The existence of cointegration between rates of different maturities is consistent with the theory of the term structure of interest rates, according to which short- and long-term rates are linked by common factors that affect the entire yield curve. These factors may include expectations regarding the future path of monetary policy, expected inflation, and general macroeconomic conditions.

Table 4: Trace statistics and p-values of the Johansen cointegration test for U.S. interest rates

Null hypothesis	Trace statistic	Critical value (5%)	p-value	Decision
$r = 0$	193.90	125.62	0.0000	Reject
$r \leq 1$	143.88	95.75	0.0000	Reject
$r \leq 2$	76.23	69.82	0.0150	Reject
$r \leq 3$	46.63	47.86	0.0671	Do not reject
$r \leq 4$	25.93	29.80	0.1334	Do not reject
$r \leq 5$	11.59	15.49	0.1800	Do not reject
$r \leq 6$	1.90	3.84	0.1677	Do not reject

2.2.2 United Kingdom

The results of the Johansen cointegration test for U.K. interest rates are presented in Table 5. The Trace statistic rejects the null hypotheses of no cointegration ($r=0$), as well as the hypotheses of at most one ($r \leq 1$) and at most two ($r \leq 2$) cointegrating vectors. However, the null hypothesis that there are at most three cointegrating vectors ($r \leq 3$) cannot be rejected at the 5% significance level. Consequently, it is concluded that the system has three cointegration relationships.

This result indicates that U.K. interest rates at different maturities share common long-run factors. Although each series is individually non-stationary, there are certain linear combinations among them that are stationary, evidencing the presence of equilibrium relationships linking the different segments of the yield curve.

The existence of cointegration suggests that short- and long-term rates do not evolve independently in the long run. On the contrary, movements observed in the different maturities tend to maintain stable relationships over time, even though temporary deviations may occur in specific periods.

Table 5: Trace statistics and p-values of the Johansen cointegration test for U.K. interest rates

Null hypothesis	Trace statistic	Critical value (5%)	p-value	Decision
$r = 0$	195.65	125.62	0.0000	Reject
$r \leq 1$	125.68	95.75	0.0001	Reject
$r \leq 2$	77.78	69.82	0.0101	Reject
$r \leq 3$	33.20	47.86	0.2932	Do not reject
$r \leq 4$	16.18	29.80	0.5957	Do not reject
$r \leq 5$	8.42	15.49	0.6871	Do not reject
$r \leq 6$	2.75	3.84	0.0973	Do not reject

The results obtained for both countries are very similar. In both cases, the Johansen test identifies three cointegrating vectors, indicating the existence of long-run equilibrium relationships between rates corresponding to different maturities. This result suggests that the rates that make up the yield curve, or term structure of interest rates, share common stochastic trends and do not evolve independently in the long run. Consequently, the different segments of the curve tend to maintain stable relationships over time, reflecting the influence of common factors that simultaneously affect short- and long-term rates.

2.3 Implied devaluations and 3-month and 10-year rate differentials

To analyze the relationship between the forward exchange market and the interest rates of both countries, implied devaluations associated with the three-month and ten-year forward contracts were constructed. The implied devaluation was calculated as the implied percentage change between the forward and spot exchange rates, according to the following expression: $DEV_t = \left(\frac{F_t - S_t}{S_t} \right) \times 100$, where (F_t) represents the forward exchange rate for the horizon considered and (S_t) the spot exchange rate observed in the same period. These series were then compared with the interest rate differentials corresponding to the same maturities. Figures 54 and 55 present the comparisons for the three-month and ten-year horizons, respectively, while Tables 6 and 7 summarize the main descriptive statistics.

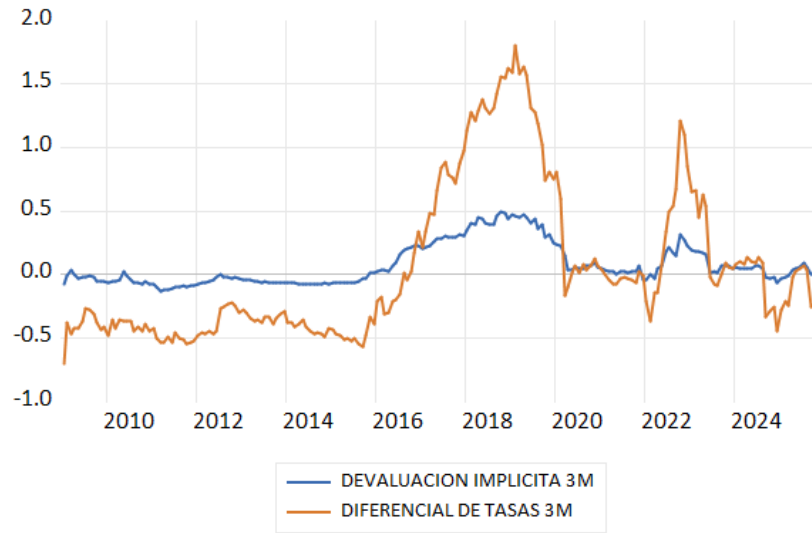


Figure 54: Implied devaluation and 3-month rate differential

Figure 54 shows that the three-month implied devaluation and the three-month rate differential exhibit a relatively similar evolution over the sample. Both series show coincident movements in different periods, particularly during the increase observed between 2015 and 2019, the subsequent decline associated with the pandemic period, and the fluctuations recorded from 2022 onward. However, the rate differential shows greater amplitude in its movements. This difference is also observed in the descriptive statistics of Table 6, where the standard deviation of the rate differential reaches 0.60, while that of the implied devaluation is approximately 0.16. Likewise, the rate differential records more extreme maximum and minimum values than the implied devaluation.

Table 6: Descriptive statistics of the implied devaluation and the 3-month rate differential

Statistic	Implied devaluation 3M	Rate differential 3M
Mean	0.0717	0.0504
Median	0.0194	-0.1688
Maximum	0.4955	1.8026
Minimum	-0.1291	-0.7038
Std. deviation	0.1630	0.6007
Skewness	1.1123	1.2400
Kurtosis	3.0943	3.4721
Jarque-Bera	42.1431	54.1694
p-value	0.0000	0.0000
Observations	204	204

The comparison for the ten-year horizon, presented in Figure 55, reveals considerably more pronounced differences. While the rate differential remains relatively bounded around zero for most of the sample, the implied devaluation exhibits much wider fluctuations, reaching values above 15% in some periods. This disparity is also reflected in the descriptive statistics of Table 7. The standard deviation of the implied devaluation is close to 5, versus a value of approximately 0.59 for the rate differential. Similarly, the maximum and minimum values observed for the implied devaluation are substantially larger than those corresponding to the rate differential.

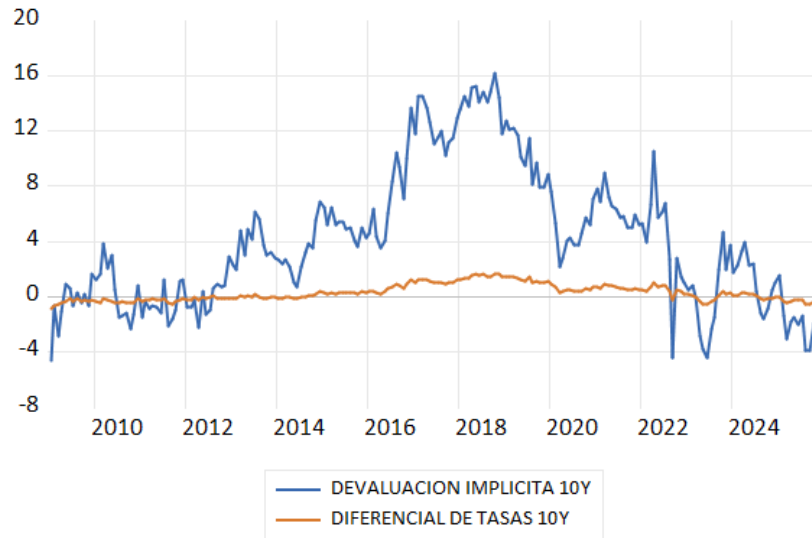


Figure 55: Implied devaluation and 10-year rate differential

From an economic standpoint, these differences can be interpreted in light of covered interest rate parity. Under this theoretical relationship, the interest rate differential between two countries should reflect the implied devaluation embedded in the forward market. The results obtained suggest that this correspondence is relatively close for the three-month horizon, where both series show a similar, though not identical, dynamic. For the ten-year horizon, by contrast, the discrepancies are considerably larger. A possible explanation is

that very long-term forward contracts incorporate greater uncertainty associated with inflation expectations, risk premia, growth prospects, and other macroeconomic factors that can affect the future evolution of the exchange rate.

Table 7: Descriptive statistics of the implied devaluation and the 10-year rate differential

Statistic	Implied devaluation 10Y	Rate differential 10Y
Mean	4.1150	0.3054
Median	3.5394	0.1974
Maximum	16.1445	1.7065
Minimum	-4.5770	-0.8637
Std. deviation	4.9879	0.5896
Skewness	0.5726	0.5711
Kurtosis	2.5281	2.3311
Jarque-Bera	13.0417	14.8914
p-value	0.0015	0.0006
Observations	204	204

Overall, the evidence indicates that the relationship between implied devaluation and the rate differential is closer for short maturities than for long maturities. This suggests that covered interest rate parity is a more accurate approximation over short-term horizons, while for longer maturities the differences between both measures tend to widen significantly.

2.4 Estimation of the Diebold-Li model via the Kalman filter, using the rate differential

In order to characterize the dynamics of the term structure of rate differentials between the United States and the United Kingdom, the Diebold-Li model was estimated via the Kalman filter. This model is a dynamic reformulation of the Nelson-Siegel model and represents the term structure using three latent factors: level, slope, and curvature. The level factor captures parallel shifts of the entire term structure, the slope factor reflects differences between short- and long-term maturities, and the curvature factor captures variations in intermediate maturities.

The estimation used as observed variables the rate differentials corresponding to the 3-month, 6-month, 12-month, 2-year, 5-year, and 10-year maturities. The latent factors were modeled as first-order autoregressive processes and estimated via the Kalman filter, which allows inferring the temporal trajectory of unobserved variables from the information contained in the observed series.

The decay parameter was fixed at $\lambda = 0.0609$. This parameter determines the shape of the model's factor loadings and, in particular, the maturity at which the curvature factor reaches its maximum influence. A value was chosen that places this maximum loading at approximately a 30-month horizon. In this way, the curvature factor mainly captures variation in intermediate maturities of the term structure, while the level and slope factors describe common movements and differences between the extremes of the curve, respectively.

The estimation results are presented in Figure 56, while the temporal evolution of the estimated factors is shown in Figure 57. To facilitate visual comparison between the series, the factors were normalized before being plotted.

Sspace: UNTITLED
Method: Maximum likelihood (BFGS / Marquardt steps)
Date: 06/10/26 Time: 01:18
Sample: 2009M01 2025M12
Included observations: 204
Estimation settings: tol= 0.00010, derivs=accurate numeric
Initial Values: C(1)=-2.38750, C(2)=0.80000, C(3)=1.48747, C(4)=0.92674,
C(5)=0.70000, C(6)=0.95881, C(10)=-4.96649, C(11)=-22.7512, C(12)=-3.70853
Convergence achieved after 83 iterations
Coefficient covariance computed using outer product of gradients

	Coefficient	Std. Error	z-Statistic	Prob.
C(1)	-0.855490	3.194374	-0.267812	0.7888
C(2)	0.916430	0.320974	2.855159	0.0043
C(3)	0.467976	0.653315	0.716310	0.4738
C(4)	0.954414	0.063219	15.09692	0.0000
C(5)	0.148087	0.146132	1.013383	0.3109
C(6)	0.988039	0.010677	92.53534	0.0000
C(10)	-4.730020	0.250048	-18.91645	0.0000
C(11)	-3.838517	0.093245	-41.16595	0.0000
C(12)	-4.237301	0.217083	-19.51926	0.0000

	Final State	Root MSE	z-Statistic	Prob.
LEVEL	-10.62108	1.251701	-8.485323	0.0000
SLOPE	10.41418	0.848998	12.26643	0.0000
CURVE	11.37680	1.040695	10.93192	0.0000

Log likelihood	-57.60348	Akaike info criterion	0.652975
Parameters	9	Schwarz criterion	0.799363
Diffuse priors	0	Hannan-Quinn criter.	0.712192

Figure 56: Estimation of the Diebold-Li model via the Kalman filter, with the rate differential as the dependent variable

In Figure 56, all three latent factors are statistically significant, with p-values below 1%. In particular, the coefficients associated with level, slope, and curvature are 0.9535, 0.9166, and 0.9127, respectively. These values indicate that all three factors show high temporal persistence, with the level factor being the most persistent. Consequently, changes affecting the term structure of differentials tend to persist over time and are not reversed immediately. This result is consistent with the nature of financial markets, where expectations about interest rates and macroeconomic conditions tend to adjust gradually.

The estimated latent factors have final values of -10.62 for level, 10.41 for slope, and 11.38 for curvature. However, since these factors evolve over time, their interpretation is more relevant through the complete trajectory presented in Figure 57 than through their point values at the end of the sample.

From an economic standpoint, the level factor captures parallel shifts of the entire term structure of differentials. An increase in this factor implies that differentials tend to increase similarly across all maturities. The slope factor, in turn, reflects differences between short- and long-term maturities, providing information about the inclination of the term structure. Finally, the curvature factor captures variations in intermediate maturities relative to the extremes, allowing changes in the shape of the curve to be captured.

Figure 57 shows that all three factors exhibit variable dynamics throughout the period analyzed. The level factor shows persistent movements and notable changes especially toward the end of the sample. The slope factor shows frequent oscillations and sign changes, reflecting changes in the relationship between short- and long-term differentials. The curvature factor, in turn, records episodes of high volatility, particularly between 2016 and 2020, suggesting relevant changes in the intermediate maturities of the term structure.

Overall, the results indicate that the dynamics of the rate differentials between the United States and the United Kingdom cannot be described solely by uniform shifts of the term structure; both slope and curvature play an important role in the observed evolution.

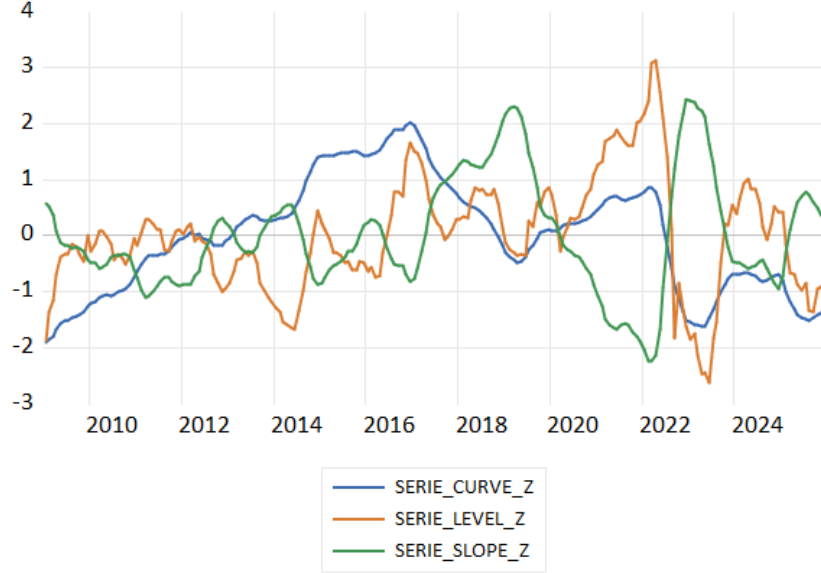


Figure 57: Temporal evolution of the estimated factors

2.5 Estimation of the Diebold-Li model via the Kalman filter, using implied devaluations

In order to analyze the term structure of implied devaluations, the Diebold-Li model was again estimated via the Kalman filter, using as observed variables the implied devaluations corresponding to the 3-month, 6-month, 12-month, 2-year, 5-year, and 10-year maturities. As in the previous section, the decay parameter was fixed at $\lambda = 0.0609$. In the Diebold-Li model, this parameter determines the shape of the factor loadings and, in particular, the maturity at which the curvature factor reaches its maximum influence. Following common practice in the literature, a value was chosen that places this maximum loading at approximately a 30-month horizon, allowing the curvature factor to mainly capture variation associated with intermediate maturities of the term structure.

The estimation results are presented in Figure 58, while the temporal evolution of the estimated factors is shown in Figure 59. To facilitate visual comparison between the series, the factors were normalized before being plotted.

Sspace: UNTITLED
Method: Maximum likelihood (BFGS / Marquardt steps)
Date: 06/10/26 Time: 05:16
Sample: 2009M01 2025M12
Included observations: 204
Failure to improve likelihood (singular hessian) after 76 iterations
Coefficient covariance computed using outer product of gradients

	Coefficient	Std. Error	z-Statistic	Prob.
C(1)	0.317604	4.400302	0.072178	0.9425
C(2)	0.994263	0.060683	16.38444	0.0000
C(3)	-0.095363	0.596587	-0.159847	0.8730
C(4)	0.998230	0.009263	107.7655	0.0000
C(5)	-0.101100	0.329405	-0.306917	0.7589
C(6)	0.998301	0.005780	172.7156	0.0000
C(11)	-3.393139	0.227904	-14.88844	0.0000
C(12)	0.647046	0.093553	6.916379	0.0000

	Final State	Root MSE	z-Statistic	Prob.
LEVEL	53.05637	3.420720	15.51029	0.0000
SLOPE	-53.03448	3.302718	-16.05783	0.0000
CURVE	-59.64153	3.589872	-16.61383	0.0000

Log likelihood	-1496.098	Akaike info criterion	14.74606
Parameters	8	Schwarz criterion	14.87618
Diffuse priors	0	Hannan-Quinn criter.	14.79870

Figure 58: Estimation of the Diebold-Li model via the Kalman filter, with implied devaluations as the dependent variable

The estimated autoregressive coefficients for the level, slope, and curvature factors are 0.9982, 0.9980, and 0.9982, respectively, all statistically significant at the 1% level. As with the model estimated on rate differentials, all three factors show high temporal persistence. However, the coefficients obtained in this case are even closer to one, suggesting particularly persistent dynamics for the factors associated with implied devaluations.

From an economic standpoint, the level factor captures parallel shifts of the entire term structure of implied devaluations, while the slope factor reflects differences between short- and long-term horizons, and the curvature factor captures variation in intermediate maturities relative to the extremes of the curve. The final estimated values for the factors are 53.06 for level, -53.03 for slope, and -59.64 for curvature, all statistically significant. Nevertheless, since these factors evolve over time, it is more relevant to analyze their complete trajectory than their point values at the end of the sample.

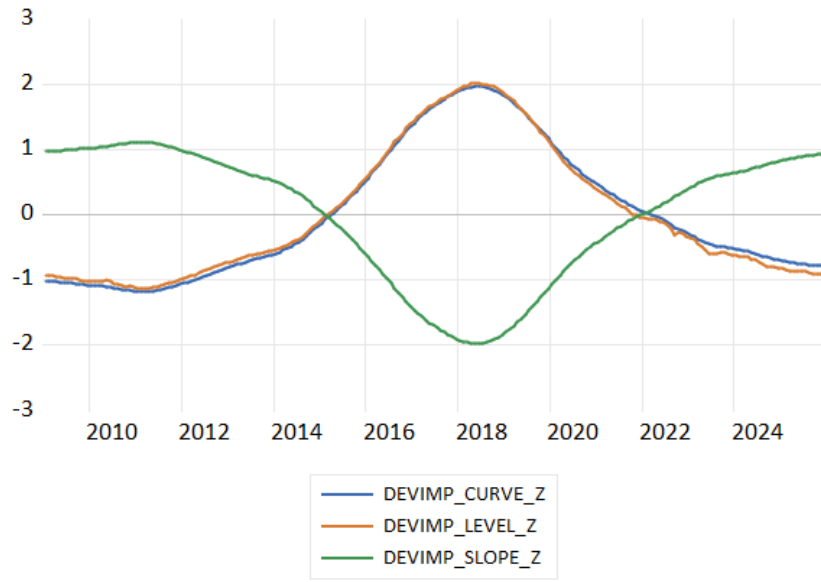


Figure 59: Temporal evolution of the estimated factors

The temporal evolution of the estimated factors is presented in Figure 59. The level and curvature factors show very similar trajectories over the sample, recording an upward trend between 2014 and 2018, followed by a gradual decline in subsequent years. By contrast, the slope factor shows approximately inverse behavior, reaching minimum values when the other two factors record their maxima. This pattern may reflect changes in the shape of the term structure of implied devaluations, where movements associated with slope partially offset variations in level and curvature. Likewise, the marked persistence of all three factors suggests that the expectations embedded in forward contracts evolve gradually over time, without abrupt changes in their dynamics.

Comparison with the results obtained in the previous section allows some differences between both estimations to be identified. In both cases, the factors show high temporal persistence and are statistically significant. However, the autoregressive coefficients estimated for implied devaluations are closer to one than those obtained for rate differentials, which could indicate greater persistence in the movements associated with the expectations embedded in forward contracts. Likewise, the trajectories of the factors estimated from implied devaluations show a smoother and more gradual evolution than that observed in the model based on rate differentials.

A Full ADF test results

A.1 United States

Null Hypothesis: US_1M has a unit root
Exogenous: Constant
Lag Length: 3 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.438800	0.5626
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_1M)
Method: Least Squares
Date: 06/07/26 Time: 08:48
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_1M(-1)	-0.010444	0.007259	-1.438800	0.1518
D(US_1M(-1))	0.102009	0.068167	1.496468	0.1361
D(US_1M(-2))	0.141923	0.068026	2.086296	0.0382
D(US_1M(-3))	0.295987	0.068664	4.310678	0.0000
C	0.020441	0.015670	1.304458	0.1936

R-squared	0.143324	Mean dependent var	0.017575
Adjusted R-squared	0.126105	S.D. dependent var	0.196275
S.E. of regression	0.183483	Akaike info criterion	-0.529188
Sum squared resid	6.699517	Schwarz criterion	-0.447861
Log likelihood	58.97717	Hannan-Quinn criter.	-0.496290
F-statistic	8.323313	Durbin-Watson stat	2.089835
Prob(F-statistic)	0.000003		

Null Hypothesis: US_6M has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.213285	0.6688
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_6M)
Method: Least Squares
Date: 06/07/26 Time: 08:50
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_6M(-1)	-0.006595	0.005435	-1.213285	0.2265
D(US_6M(-1))	0.378781	0.069188	5.474636	0.0000
D(US_6M(-2))	0.203989	0.067681	3.013991	0.0029
C	0.016530	0.012292	1.344794	0.1802

R-squared	0.259466	Mean dependent var	0.016386
Adjusted R-squared	0.248358	S.D. dependent var	0.159896
S.E. of regression	0.138626	Akaike info criterion	-1.094668
Sum squared resid	3.843407	Schwarz criterion	-1.029607
Log likelihood	115.6561	Hannan-Quinn criter.	-1.068350
F-statistic	23.35840	Durbin-Watson stat	2.069283
Prob(F-statistic)	0.000000		

Null Hypothesis: US_3M has a unit root
Exogenous: Constant
Lag Length: 3 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.649329	0.4555
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_3M)
Method: Least Squares
Date: 06/07/26 Time: 08:49
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_3M(-1)	-0.008605	0.005217	-1.649329	0.1007
D(US_3M(-1))	0.399163	0.067625	5.902612	0.0000
D(US_3M(-2))	0.029020	0.072758	0.398862	0.6904
D(US_3M(-3))	0.267133	0.066605	4.010681	0.0001
C	0.016916	0.011512	1.469448	0.1433

R-squared	0.322057	Mean dependent var	0.017403
Adjusted R-squared	0.308430	S.D. dependent var	0.159797
S.E. of regression	0.132888	Akaike info criterion	-1.174414
Sum squared resid	3.514192	Schwarz criterion	-1.093087
Log likelihood	124.7902	Hannan-Quinn criter.	-1.141516
F-statistic	23.63373	Durbin-Watson stat	2.043929
Prob(F-statistic)	0.000000		

Null Hypothesis: US_12M has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.129130	0.7043
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_12M)
Method: Least Squares
Date: 06/07/26 Time: 08:48
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_12M(-1)	-0.007390	0.006545	-1.129130	0.2602
D(US_12M(-1))	0.255813	0.068018	3.760943	0.0002
D(US_12M(-2))	0.197688	0.067532	2.927319	0.0038
C	0.020251	0.014470	1.399512	0.1632

R-squared	0.146879	Mean dependent var	0.015341
Adjusted R-squared	0.134082	S.D. dependent var	0.169857
S.E. of regression	0.158060	Akaike info criterion	-0.832266
Sum squared resid	4.996618	Schwarz criterion	-0.767205
Log likelihood	88.89114	Hannan-Quinn criter.	-0.805948
F-statistic	11.47775	Durbin-Watson stat	1.957608
Prob(F-statistic)	0.000001		

Figure 60: ADF test results for the short-term U.S. rates.

Null Hypothesis: US_2Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.650833	0.8551
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_2Y)
Method: Least Squares
Date: 06/07/26 Time: 08:49
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_2Y(-1)	-0.006309	0.009694	-0.650833	0.5159
C	0.023142	0.021059	1.098893	0.2731
R-squared	0.002093	Mean dependent var	0.013278	
Adjusted R-squared	-0.002848	S.D. dependent var	0.208543	
S.E. of regression	0.208840	Akaike info criterion	-0.284742	
Sum squared resid	8.810051	Schwarz criterion	-0.252211	
Log likelihood	31.04367	Hannan-Quinn criter.	-0.271583	
F-statistic	0.423583	Durbin-Watson stat	1.780169	
Prob(F-statistic)	0.515894			

Null Hypothesis: US_5Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.215902	0.6677
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_5Y)
Method: Least Squares
Date: 06/07/26 Time: 08:50
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_5Y(-1)	-0.018245	0.015005	-1.215902	0.2254
C	0.047936	0.035255	1.359686	0.1754
R-squared	0.007266	Mean dependent var	0.010668	
Adjusted R-squared	0.002351	S.D. dependent var	0.249112	
S.E. of regression	0.248819	Akaike info criterion	0.065570	
Sum squared resid	12.50596	Schwarz criterion	0.098101	
Log likelihood	-4.688150	Hannan-Quinn criter.	0.078729	
F-statistic	1.478419	Durbin-Watson stat	1.959316	
Prob(F-statistic)	0.225441			

Null Hypothesis: US_10Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.453557	0.5552
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(US_10Y)
Method: Least Squares
Date: 06/07/26 Time: 08:47
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
US_10Y(-1)	-0.025294	0.017401	-1.453557	0.1476
C	0.075672	0.048612	1.556636	0.1211
R-squared	0.010351	Mean dependent var	0.009582	
Adjusted R-squared	0.005452	S.D. dependent var	0.246342	
S.E. of regression	0.245669	Akaike info criterion	0.040096	
Sum squared resid	12.19141	Schwarz criterion	0.072627	
Log likelihood	-2.089808	Hannan-Quinn criter.	0.053255	
F-statistic	2.112827	Durbin-Watson stat	1.837867	
Prob(F-statistic)	0.147621			

Figure 61: ADF test results for the long-term U.S. rates.

A.2 United Kingdom

Null Hypothesis: UK_1M has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.191649	0.6782
Test critical values:		
1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_1M)
Method: Least Squares
Date: 06/07/26 Time: 08:55
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_1M(-1)	-0.019250	0.016154	-1.191649	0.2348
C	0.034680	0.032067	1.081470	0.2808
R-squared	0.006981	Mean dependent var	0.012034	
Adjusted R-squared	0.002065	S.D. dependent var	0.369304	
S.E. of regression	0.368922	Akaike info criterion	0.853294	
Sum squared resid	27.49293	Schwarz criterion	0.885824	
Log likelihood	-85.03595	Hannan-Quinn criter.	0.866453	
F-statistic	1.420028	Durbin-Watson stat	2.475501	
Prob(F-statistic)	0.234797			

Null Hypothesis: UK_6M has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.455216	0.8958
Test critical values:		
1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_6M)
Method: Least Squares
Date: 06/07/26 Time: 08:57
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_6M(-1)	-0.003005	0.006601	-0.455216	0.6494
D(UK_6M(-1))	0.137214	0.069383	1.977634	0.0493
D(UK_6M(-2))	0.167544	0.054606	3.068229	0.0025
C	0.015452	0.014055	1.099426	0.2729
R-squared	0.074749	Mean dependent var	0.014108	
Adjusted R-squared	0.060870	S.D. dependent var	0.162742	
S.E. of regression	0.157711	Akaike info criterion	-0.836688	
Sum squared resid	4.974572	Schwarz criterion	-0.771627	
Log likelihood	89.34217	Hannan-Quinn criter.	-0.810370	
F-statistic	5.385816	Durbin-Watson stat	2.054066	
Prob(F-statistic)	0.001385			

Null Hypothesis: UK_3M has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.921836	0.7798
Test critical values:		
1% level	-3.462737	
5% level	-2.875680	
10% level	-2.574385	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_3M)
Method: Least Squares
Date: 06/07/26 Time: 08:56
Sample (adjusted): 2009M03 2025M12
Included observations: 202 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_3M(-1)	-0.003840	0.004166	-0.921836	0.3577
D(UK_3M(-1))	0.320946	0.066052	4.859001	0.0000
D(UK_3M(-2))	0.342811	0.066105	5.185831	0.0000
C	0.011303	0.008652	1.306341	0.1930
R-squared	0.336975	Mean dependent var	0.016158	
Adjusted R-squared	0.326929	S.D. dependent var	0.119580	
S.E. of regression	0.098105	Akaike info criterion	-1.785960	
Sum squared resid	1.905657	Schwarz criterion	-1.720450	
Log likelihood	184.3819	Hannan-Quinn criter.	-1.759454	
F-statistic	33.54373	Durbin-Watson stat	2.026667	
Prob(F-statistic)	0.000000			

Null Hypothesis: UK_12M has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.141533	0.9421
Test critical values:		
1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_12M)
Method: Least Squares
Date: 06/07/26 Time: 08:54
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_12M(-1)	-0.001183	0.008356	-0.141533	0.8876
C	0.017146	0.016570	1.034781	0.3020
R-squared	0.000099	Mean dependent var	0.015711	
Adjusted R-squared	-0.004851	S.D. dependent var	0.186690	
S.E. of regression	0.187143	Akaike info criterion	-0.504136	
Sum squared resid	7.074518	Schwarz criterion	-0.471605	
Log likelihood	53.42184	Hannan-Quinn criter.	-0.490977	
F-statistic	0.020032	Durbin-Watson stat	1.862589	
Prob(F-statistic)	0.887590			

Figure 62: ADF test results for the short-term U.K. rates.

Null Hypothesis: UK_2Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.449109	0.8969
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_2Y)
Method: Least Squares
Date: 06/07/26 Time: 08:55
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_2Y(-1)	-0.004831	0.010756	-0.449109	0.6538
C	0.019298	0.020834	0.926271	0.3554
R-squared	0.000998	Mean dependent var	0.013176	
Adjusted R-squared	-0.003948	S.D. dependent var	0.224609	
S.E. of regression	0.225051	Akaike info criterion	-0.135220	
Sum squared resid	10.23093	Schwarz criterion	-0.102689	
Log likelihood	15.79240	Hannan-Quinn criter.	-0.122060	
F-statistic	0.201699	Durbin-Watson stat	1.924377	
Prob(F-statistic)	0.653834			

Null Hypothesis: UK_5Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.960876	0.7669
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_5Y)
Method: Least Squares
Date: 06/07/26 Time: 08:56
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_5Y(-1)	-0.013288	0.013829	-0.960876	0.3378
C	0.030286	0.030048	1.007926	0.3147
R-squared	0.004550	Mean dependent var	0.007333	
Adjusted R-squared	-0.000378	S.D. dependent var	0.260305	
S.E. of regression	0.260354	Akaike info criterion	0.156206	
Sum squared resid	13.69240	Schwarz criterion	0.188737	
Log likelihood	-13.93302	Hannan-Quinn criter.	0.169365	
F-statistic	0.923283	Durbin-Watson stat	2.023122	
Prob(F-statistic)	0.337763			

Null Hypothesis: UK_10Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.944770	0.7723
Test critical values: 1% level	-3.462412	
5% level	-2.875538	
10% level	-2.574309	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(UK_10Y)
Method: Least Squares
Date: 06/07/26 Time: 08:53
Sample: 2009M01 2025M12
Included observations: 204

Variable	Coefficient	Std. Error	t-Statistic	Prob.
UK_10Y(-1)	-0.012888	0.013641	-0.944770	0.3459
C	0.036921	0.036221	1.019341	0.3093
R-squared	0.004399	Mean dependent var	0.007152	
Adjusted R-squared	-0.000529	S.D. dependent var	0.255067	
S.E. of regression	0.255134	Akaike info criterion	0.115703	
Sum squared resid	13.14890	Schwarz criterion	0.148234	
Log likelihood	-9.801742	Hannan-Quinn criter.	0.128863	
F-statistic	0.892591	Durbin-Watson stat	2.046157	
Prob(F-statistic)	0.345905			

Figure 63: ADF test results for the long-term U.K. rates.

A.3 Rate differentials

Null Hypothesis: DIF_1M has a unit root
Exogenous: Constant
Lag Length: 4 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.747258	0.4059
Test critical values: 1% level	-3.463235	
5% level	-2.875898	
10% level	-2.574501	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_1M)
Method: Least Squares
Date: 06/07/26 Time: 08:59
Sample (adjusted): 2009M06 2025M12
Included observations: 199 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_1M(-1)	-0.074623	0.042708	-1.747258	0.0822
D(DIF_1M(-1))	-0.480012	0.075973	-6.318192	0.0000
D(DIF_1M(-2))	-0.373251	0.078866	-4.732714	0.0000
D(DIF_1M(-3))	-0.321918	0.077607	-4.148051	0.0001
D(DIF_1M(-4))	-0.276058	0.072658	-3.799428	0.0002
C	0.012416	0.024469	0.507409	0.6124

R-squared	0.274608	Mean dependent var	-6.73E-05
Adjusted R-squared	0.255816	S.D. dependent var	0.394416
S.E. of regression	0.340248	Akaike info criterion	0.711401
Sum squared resid	22.34332	Schwarz criterion	0.810697
Log likelihood	-64.78442	Hannan-Quinn criter.	0.751589
F-statistic	14.61263	Durbin-Watson stat	1.928666
Prob(F-statistic)	0.000000		

Null Hypothesis: DIF_6M has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.937281	0.3147
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_6M)
Method: Least Squares
Date: 06/07/26 Time: 09:03
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_6M(-1)	-0.034683	0.017903	-1.937281	0.0541
C	0.004102	0.011076	0.370348	0.7115

R-squared	0.018330	Mean dependent var	0.001505
Adjusted R-squared	0.013446	S.D. dependent var	0.157714
S.E. of regression	0.156650	Akaike info criterion	-0.859805
Sum squared resid	4.932371	Schwarz criterion	-0.827162
Log likelihood	89.27017	Hannan-Quinn criter.	-0.846599
F-statistic	3.753057	Durbin-Watson stat	2.288538
Prob(F-statistic)	0.054113		

Null Hypothesis: DIF_3M has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.716549	0.4214
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_3M)
Method: Least Squares
Date: 06/07/26 Time: 09:03
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_3M(-1)	-0.025214	0.014689	-1.716549	0.0876
C	0.003458	0.008851	0.390747	0.6964

R-squared	0.014448	Mean dependent var	0.002149
Adjusted R-squared	0.009544	S.D. dependent var	0.126240
S.E. of regression	0.125636	Akaike info criterion	-1.301053
Sum squared resid	3.172663	Schwarz criterion	-1.268411
Log likelihood	134.0569	Hannan-Quinn criter.	-1.287847
F-statistic	2.946540	Durbin-Watson stat	1.738419
Prob(F-statistic)	0.087602		

Null Hypothesis: DIF_12M has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.241399	0.1925
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_12M)
Method: Least Squares
Date: 06/07/26 Time: 08:59
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_12M(-1)	-0.046199	0.020612	-2.241399	0.0261
C	0.011860	0.013945	0.850531	0.3960

R-squared	0.024385	Mean dependent var	0.001522
Adjusted R-squared	0.019531	S.D. dependent var	0.189356
S.E. of regression	0.187498	Akaike info criterion	-0.500298
Sum squared resid	7.066226	Schwarz criterion	-0.467655
Log likelihood	52.78024	Hannan-Quinn criter.	-0.487092
F-statistic	5.023871	Durbin-Watson stat	2.077126
Prob(F-statistic)	0.026094		

Figure 64: ADF test results for the short-term interest rate differentials.

Null Hypothesis: DIF_2Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.258779	0.1865
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_2Y)
Method: Least Squares
Date: 06/07/26 Time: 09:00
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_2Y(-1)	-0.046930	0.020777	-2.258779	0.0250
C	0.015469	0.014775	1.046992	0.2964
R-squared	0.024755	Mean dependent var	0.001437	
Adjusted R-squared	0.019903	S.D. dependent var	0.192929	
S.E. of regression	0.190999	Akaike info criterion	-0.463294	
Sum squared resid	7.332600	Schwarz criterion	-0.430652	
Log likelihood	49.02437	Hannan-Quinn criter.	-0.450088	
F-statistic	5.102081	Durbin-Watson stat	2.125489	
Prob(F-statistic)	0.024971			

Null Hypothesis: DIF_5Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.531382	0.1096
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_5Y)
Method: Least Squares
Date: 06/07/26 Time: 09:03
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_5Y(-1)	-0.051812	0.020468	-2.531382	0.0121
C	0.020469	0.014782	1.384798	0.1676
R-squared	0.030895	Mean dependent var	0.003828	
Adjusted R-squared	0.026074	S.D. dependent var	0.191138	
S.E. of regression	0.188630	Akaike info criterion	-0.488258	
Sum squared resid	7.151818	Schwarz criterion	-0.455615	
Log likelihood	51.55817	Hannan-Quinn criter.	-0.475052	
F-statistic	6.407893	Durbin-Watson stat	2.240839	
Prob(F-statistic)	0.012127			

Null Hypothesis: DIF_10Y has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=12)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.368352	0.1521
Test critical values: 1% level	-3.462574	
5% level	-2.875608	
10% level	-2.574346	

*Mackinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIF_10Y)
Method: Least Squares
Date: 06/07/26 Time: 08:58
Sample (adjusted): 2009M02 2025M12
Included observations: 203 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIF_10Y(-1)	-0.045960	0.019406	-2.368352	0.0188
C	0.016894	0.012885	1.311162	0.1913
R-squared	0.027148	Mean dependent var	0.002718	
Adjusted R-squared	0.022308	S.D. dependent var	0.164412	
S.E. of regression	0.162568	Akaike info criterion	-0.785636	
Sum squared resid	5.312109	Schwarz criterion	-0.752993	
Log likelihood	81.74202	Hannan-Quinn criter.	-0.772430	
F-statistic	5.609090	Durbin-Watson stat	2.217381	
Prob(F-statistic)	0.018816			

Figure 65: ADF test results for the long-term interest rate differentials.

B Johansen test results

Date: 06/07/26 Time: 09:32
Sample: 2009M01 2025M12
Included observations: 204
Trend assumption: Linear deterministic trend
Series: US_10Y US_12M US_1M US_2Y US_3M US_5Y US_6M
Lags interval (in first differences): 1 to 4

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.254894	193.9021	125.6154	0.0000
At most 1 *	0.246187	133.8795	95.75366	0.0000
At most 2 *	0.135051	76.22680	69.81889	0.0141
At most 3	0.121620	46.62955	47.85613	0.0649
At most 4	0.061938	20.17575	29.79707	0.4110
At most 5	0.020638	7.132142	15.49471	0.5623
At most 6	0.014009	2.878002	3.841465	0.0898

Trace test indicates 3 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Maximum Eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None *	0.254894	60.02261	46.23142	0.0010
At most 1 *	0.246187	57.65267	40.07757	0.0002
At most 2	0.135051	29.59725	33.87687	0.1491
At most 3	0.121620	26.45380	27.58434	0.0692
At most 4	0.061938	13.04361	21.13162	0.4482
At most 5	0.020638	4.254140	14.26460	0.8316
At most 6	0.014009	2.878002	3.841465	0.0898

Max-eigenvalue test indicates 2 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Table 8: Johansen test results for the United States

Date: 06/07/26 Time: 20:29
Sample (adjusted): 2009M05 2025M12
Included observations: 200 after adjustments
Trend assumption: Linear deterministic trend
Series: UK_10Y UK_12M UK_2Y UK_1M UK_3M UK_5Y UK_6M
Lags interval (in first differences): 1 to 4

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.291025	195.6477	125.6154	0.0000
At most 1 *	0.217627	126.8608	95.75366	0.0001
At most 2 *	0.179520	77.77612	69.81889	0.0101
At most 3	0.104272	38.20286	47.85613	0.2932
At most 4	0.047994	16.17916	29.79707	0.6997
At most 5	0.017804	6.342394	15.49471	0.6551
At most 6	0.013654	2.749546	3.841465	0.0973

Trace test indicates 3 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Maximum Eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None *	0.291025	68.78694	46.23142	0.0001
At most 1 *	0.217627	49.08463	40.07757	0.0038
At most 2 *	0.179520	39.57326	33.87687	0.0094
At most 3	0.104272	22.02369	27.58434	0.2191
At most 4	0.047994	9.836771	21.13162	0.7598
At most 5	0.017804	3.592848	14.26460	0.8996
At most 6	0.013654	2.749546	3.841465	0.0973

Max-eigenvalue test indicates 3 cointegrating eqn(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Table 9: Johansen test results for the United Kingdom